Particle methods with deforming basis functions and field interpolation

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The point vortex method: a venerable technique...

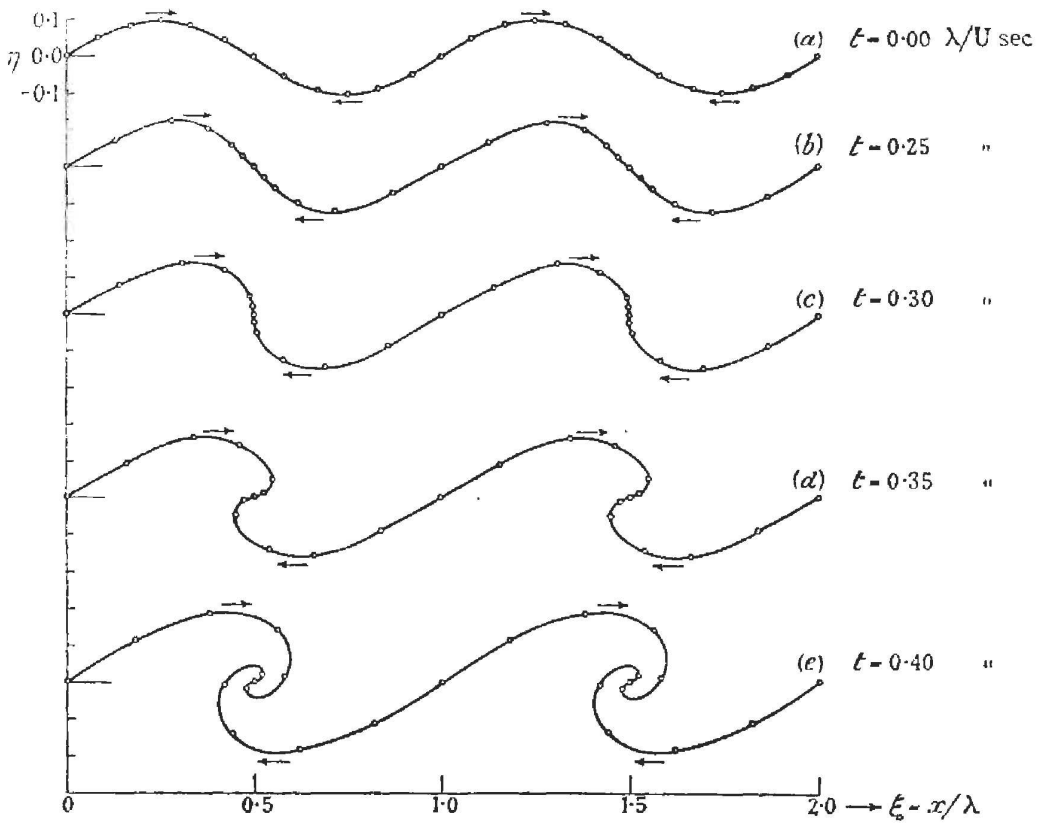


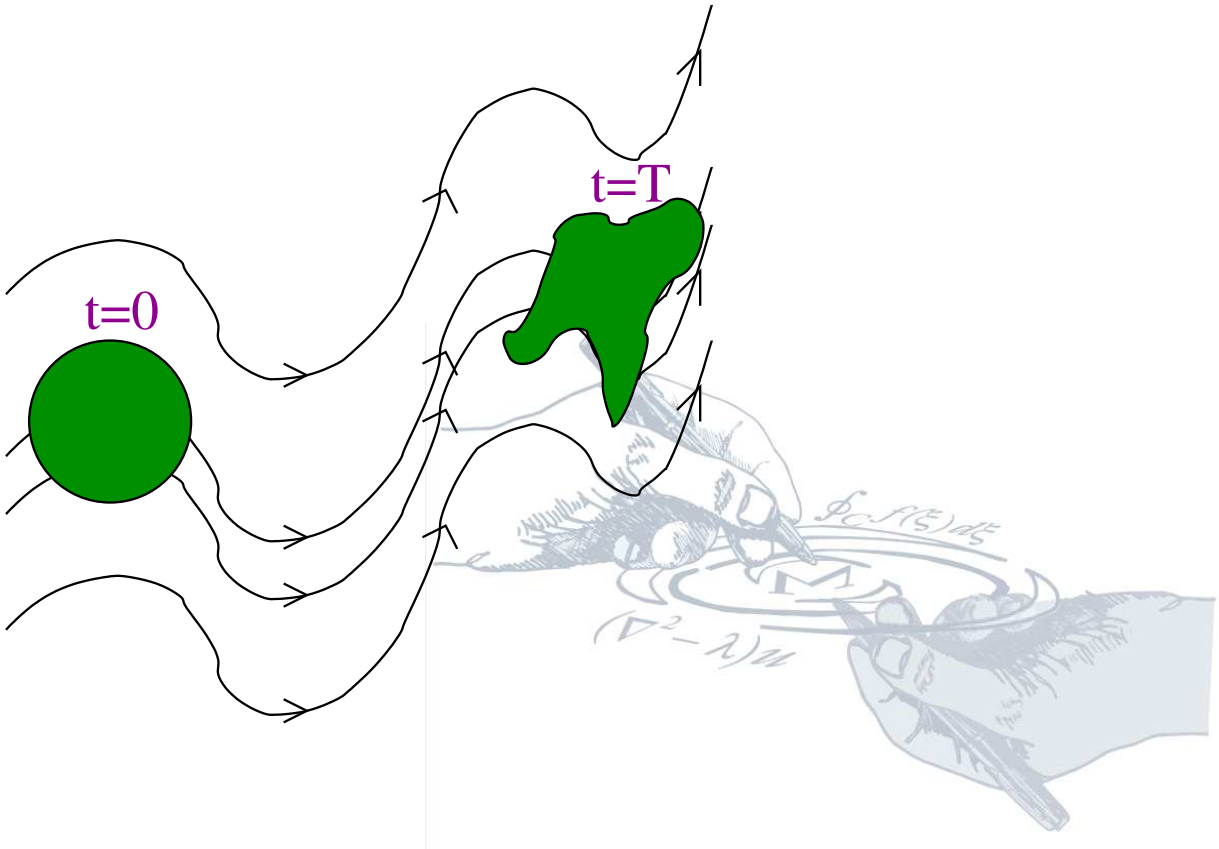
FIG. 4.

L. Rosenhead, "The point vortex approximation of a vortex sheet" *Proc. Roy. Soc.*, 134, 1932.

Particle methods

Computationally, if we know the initial distribution of ρ or ω, we can use Φ_t to follow it forward in time.

$$\frac{d\vec{x}}{dt} = \vec{u}(\vec{x}, t)$$
$$\vec{x}(0) = \vec{y}$$



Particle methods

Represent ρ as a collection of N particles, and follow them forward in time. Thus, we reduce a PDE to a $d \times N$ dimensional system of ODEs.

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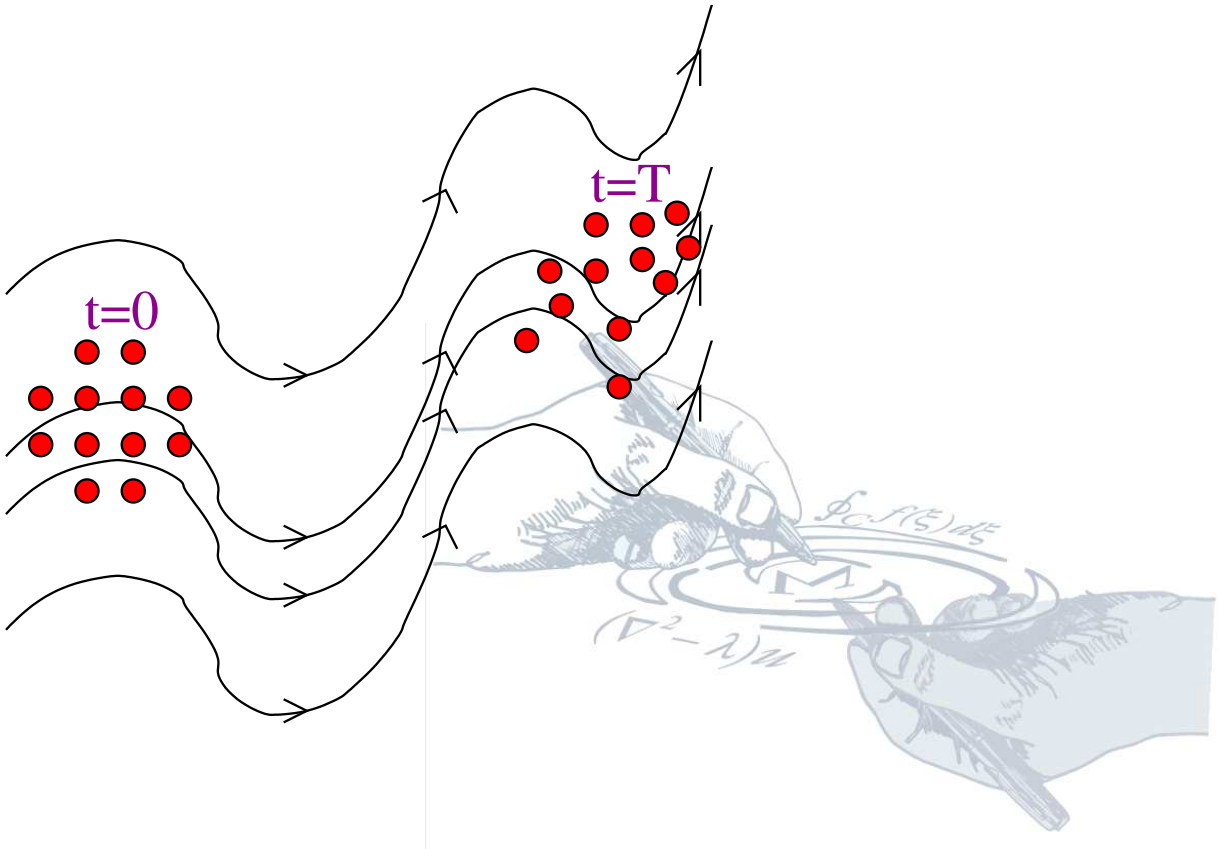

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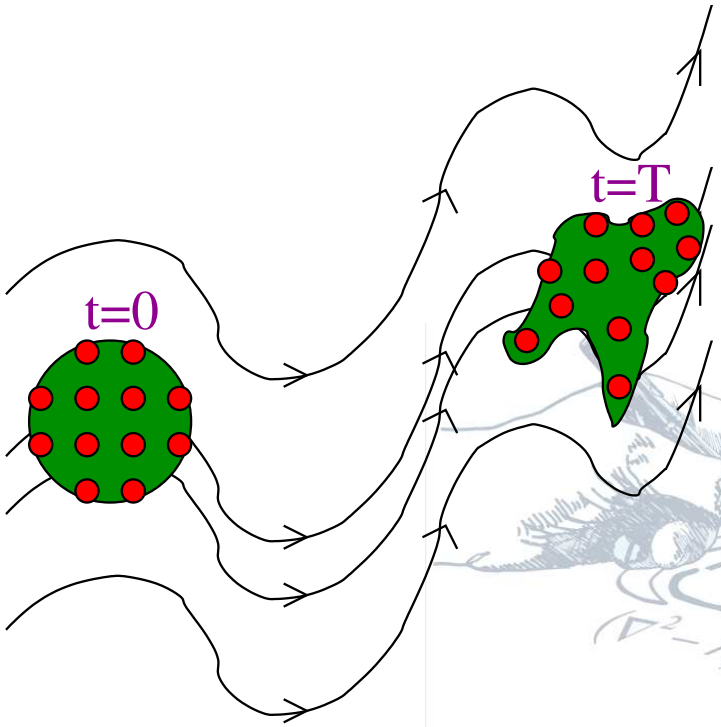

The big picture

- Our efforts need to lead to a high performance method (accurate and scalable).
- Natural adaptivity: Particle methods ought to excel for processes that can be described entirely along characteristics. (Convection)
- The leading edge has always been capturing spatial interactions. (Diffusion, visco-elastic effects, no-slip boundary conditions)
- One approach is to “build up” a method with better dynamics.

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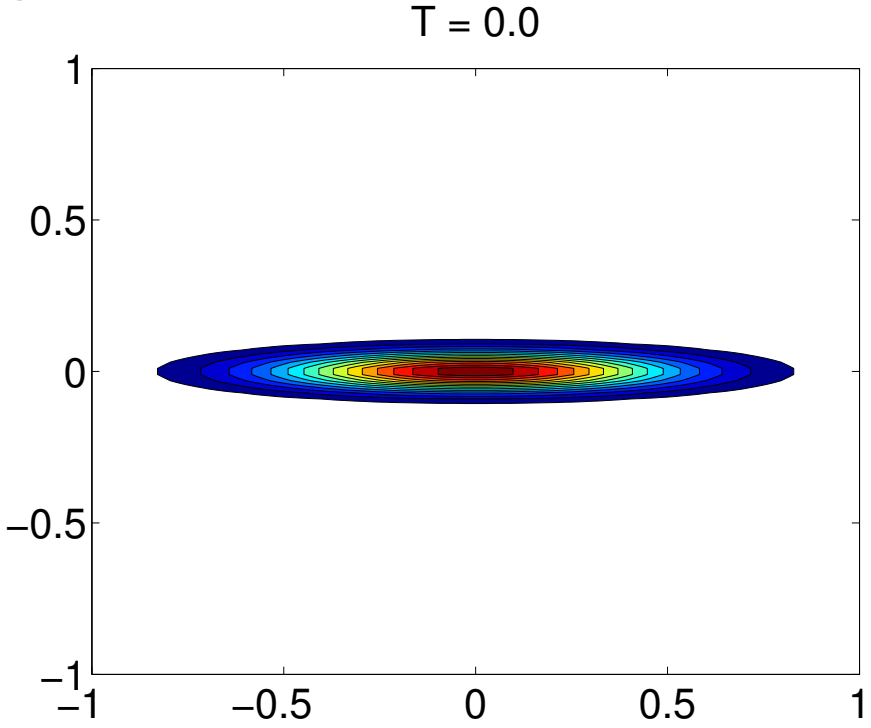
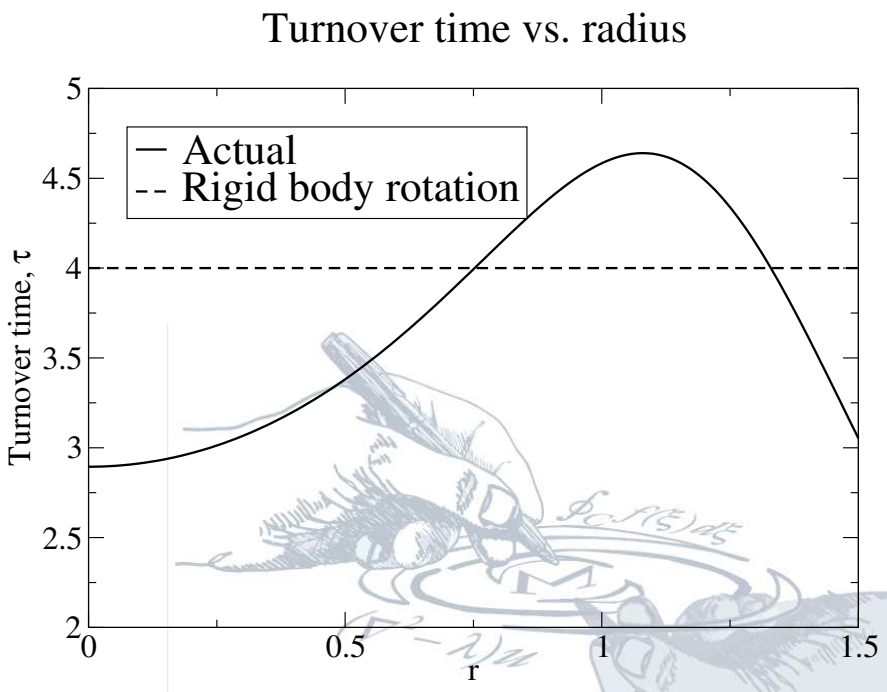
Example: Mixing in a differentially rotating flow.

Advantage: Adaptive resolution of fine scales and high gradients. Particle methods can capture diffusion exactly.





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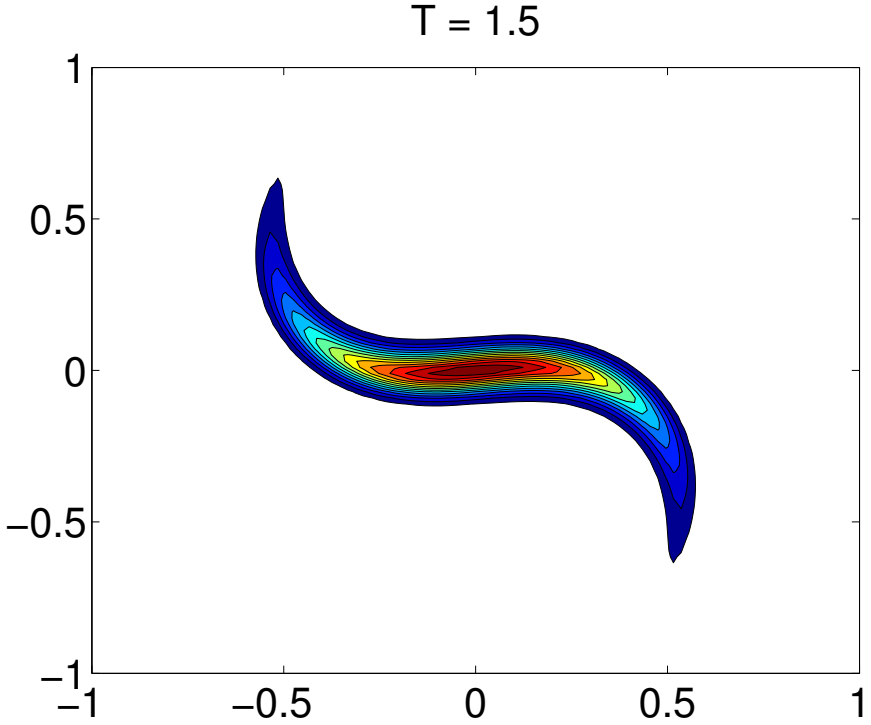
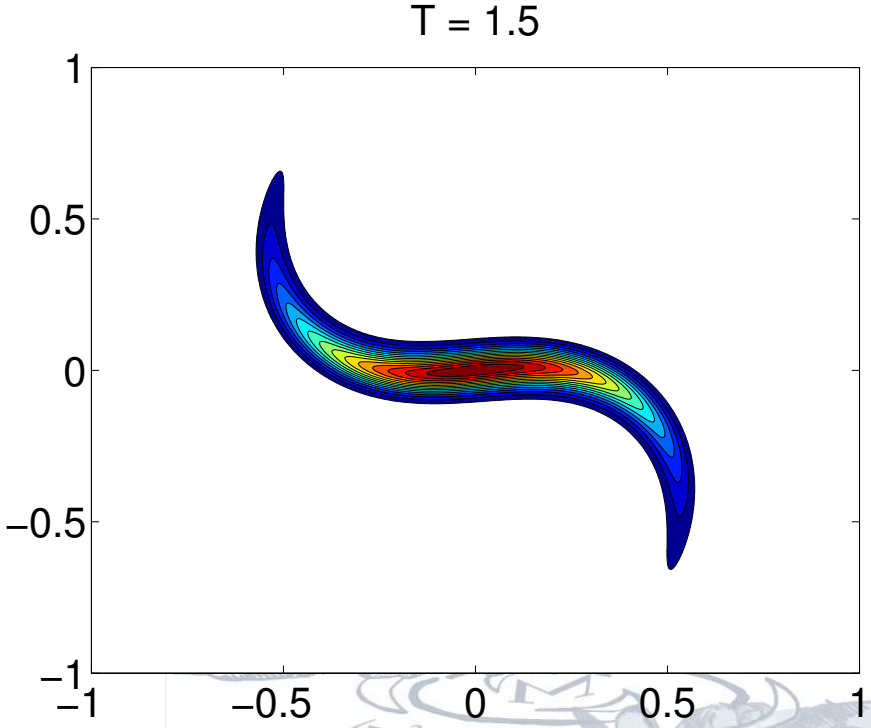


$\mathrm{Pe} = 10^4$



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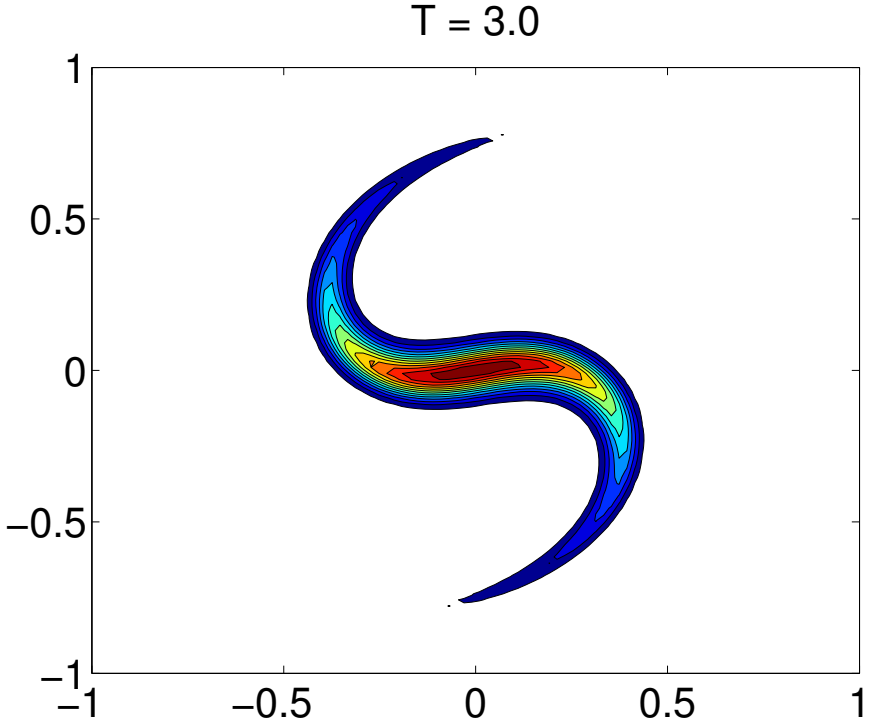

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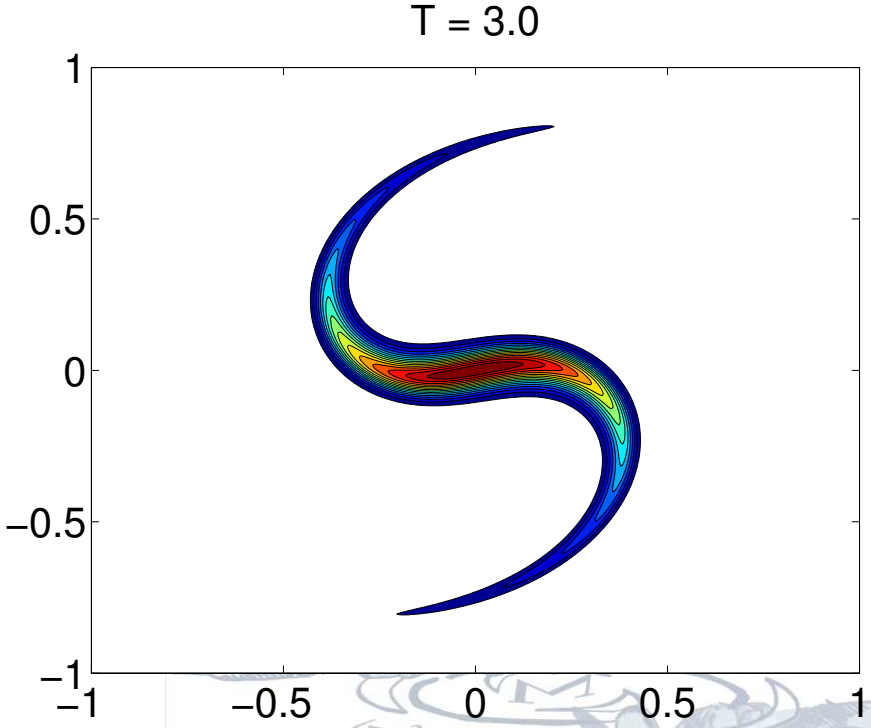

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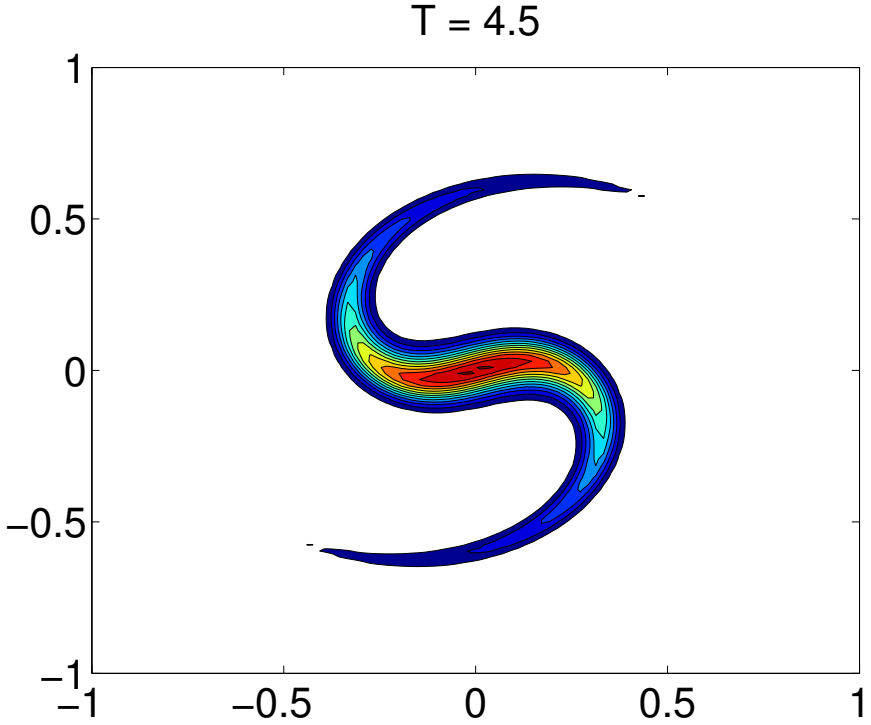

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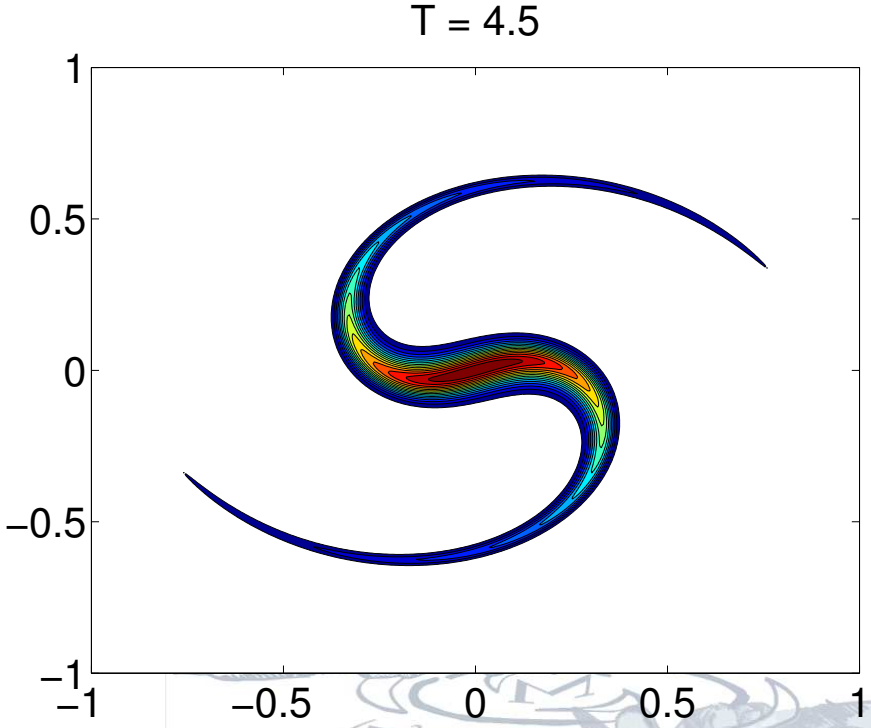

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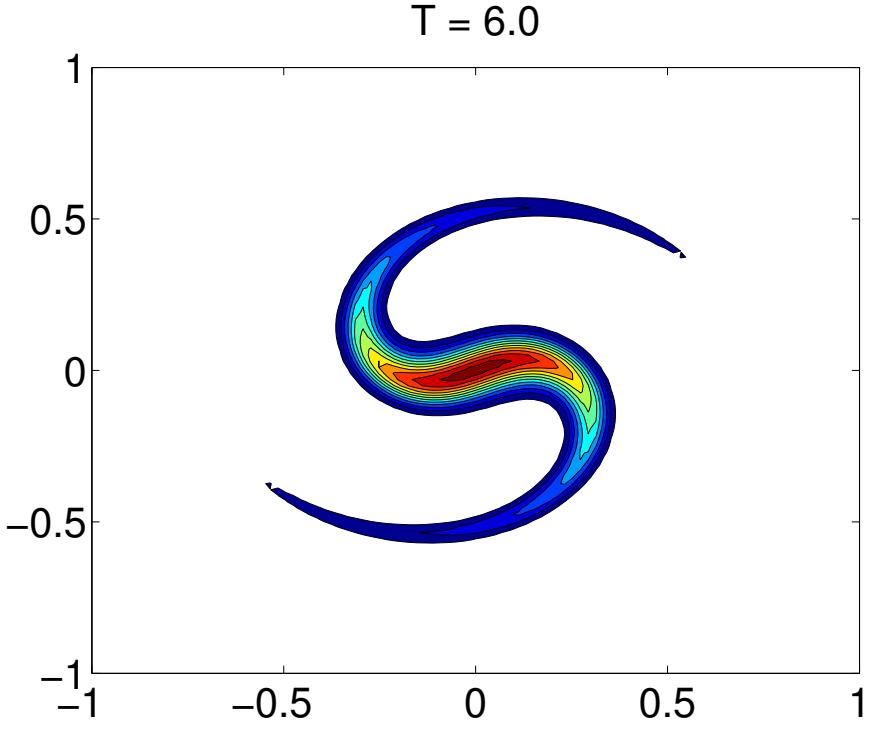

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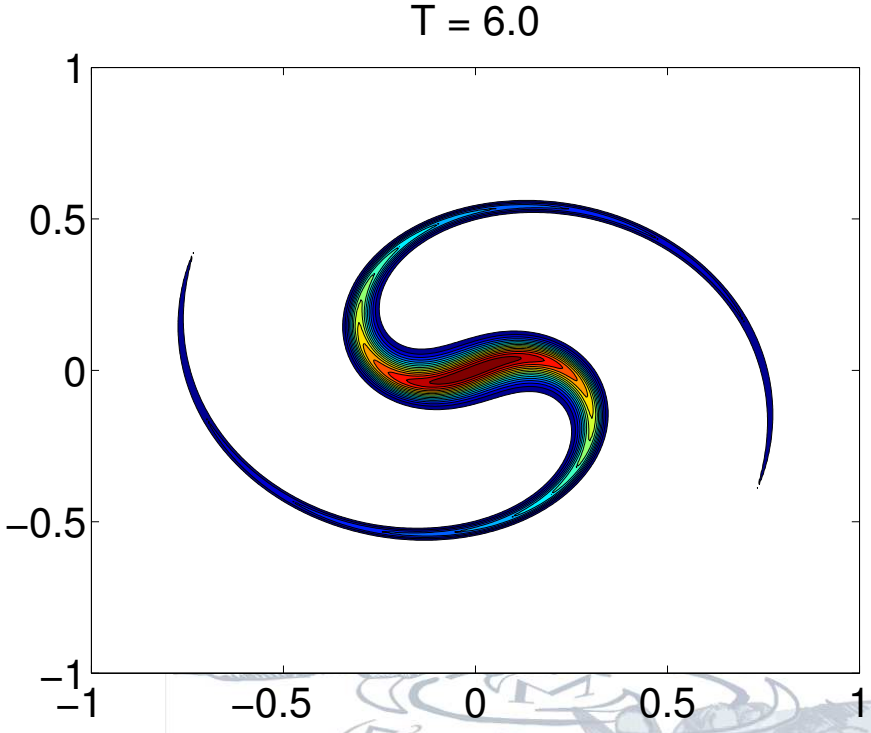

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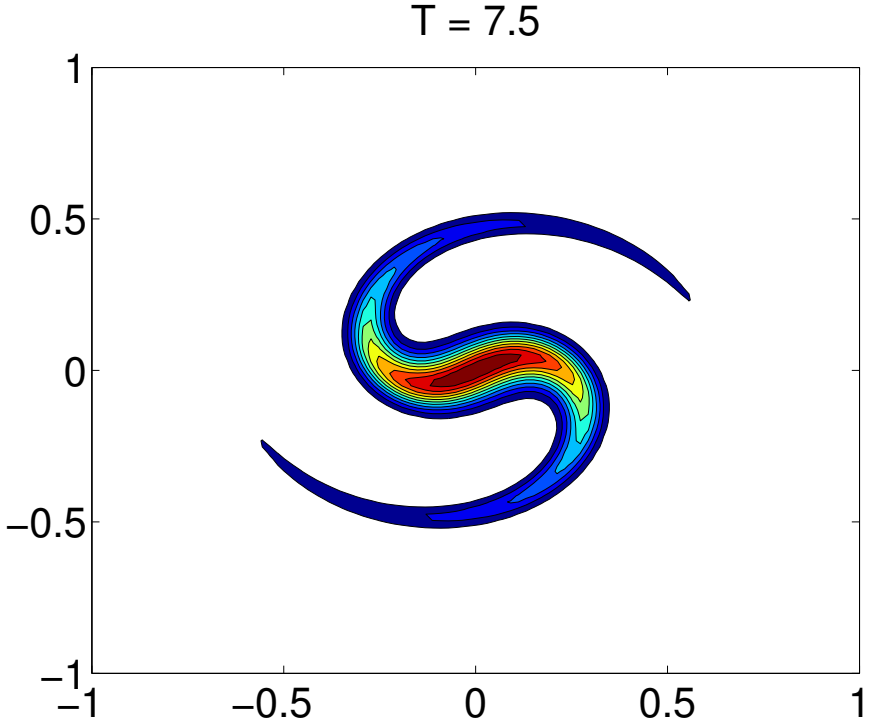

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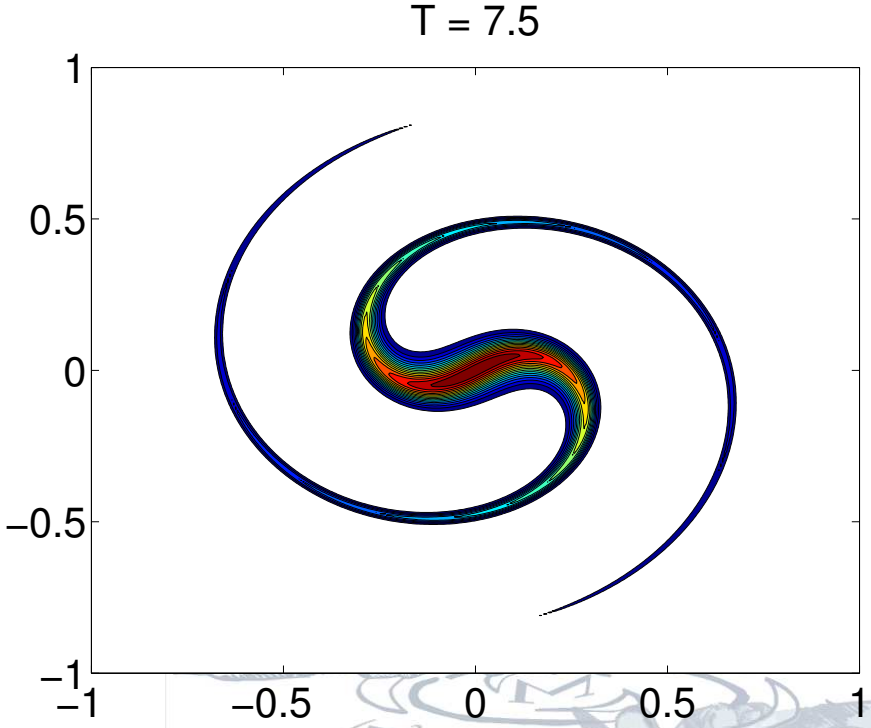

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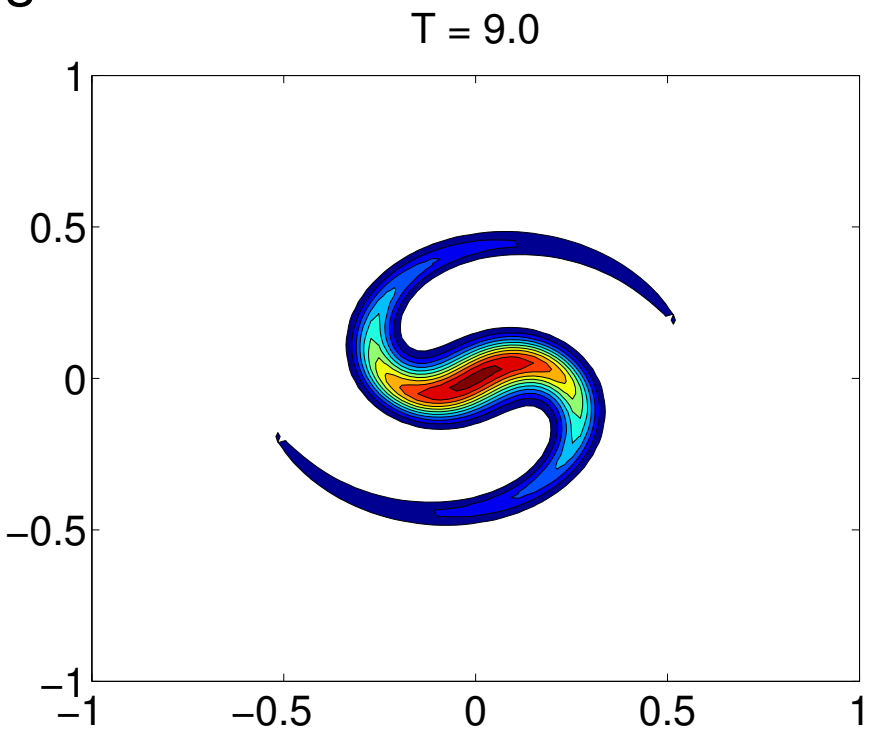

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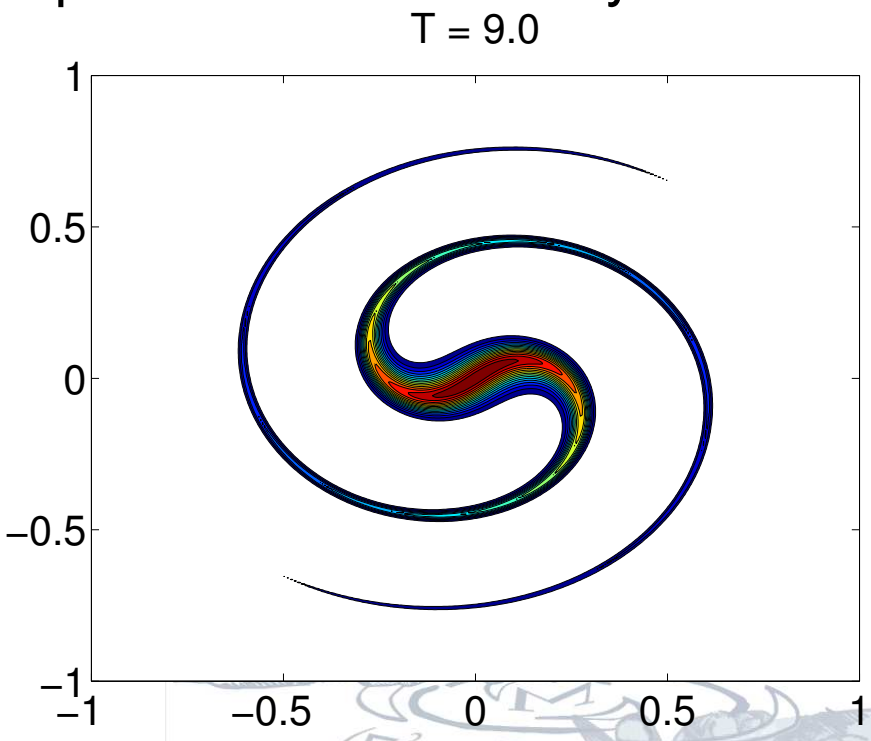

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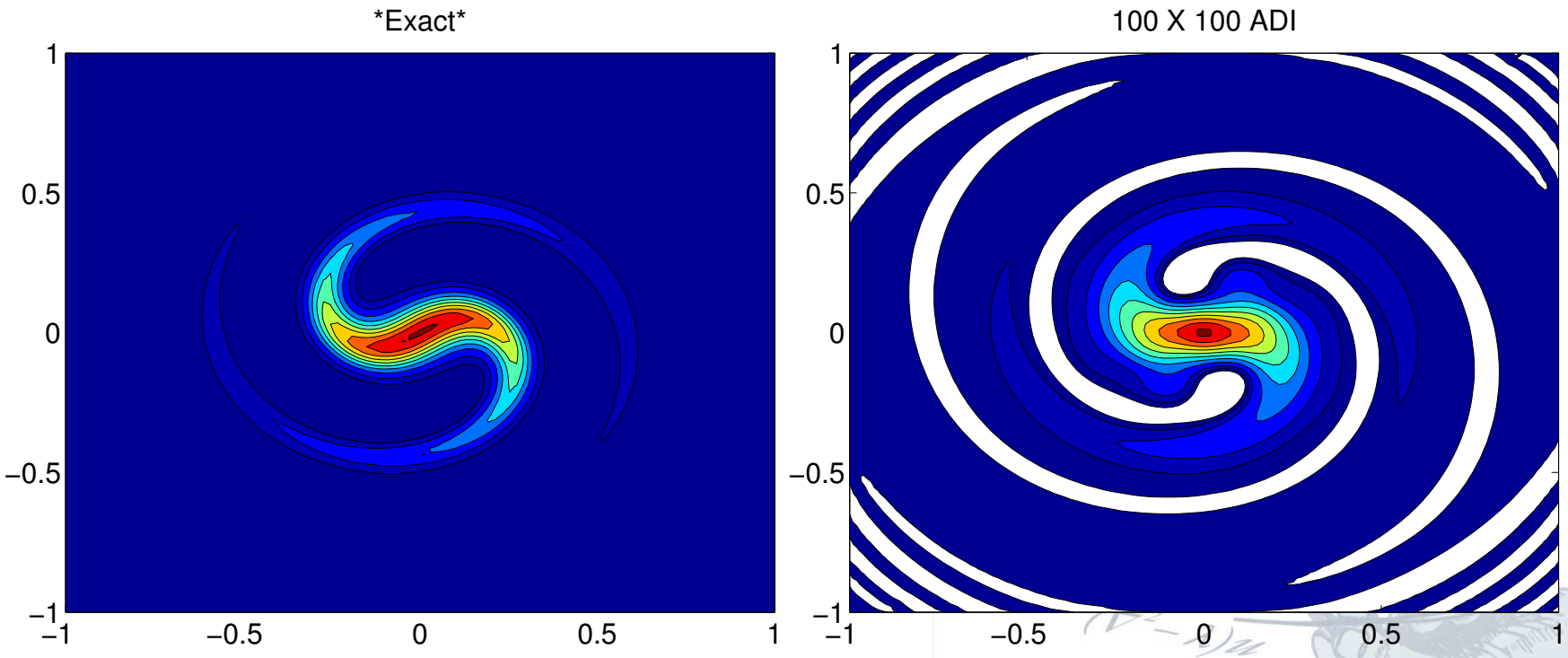
What about other general schemes?



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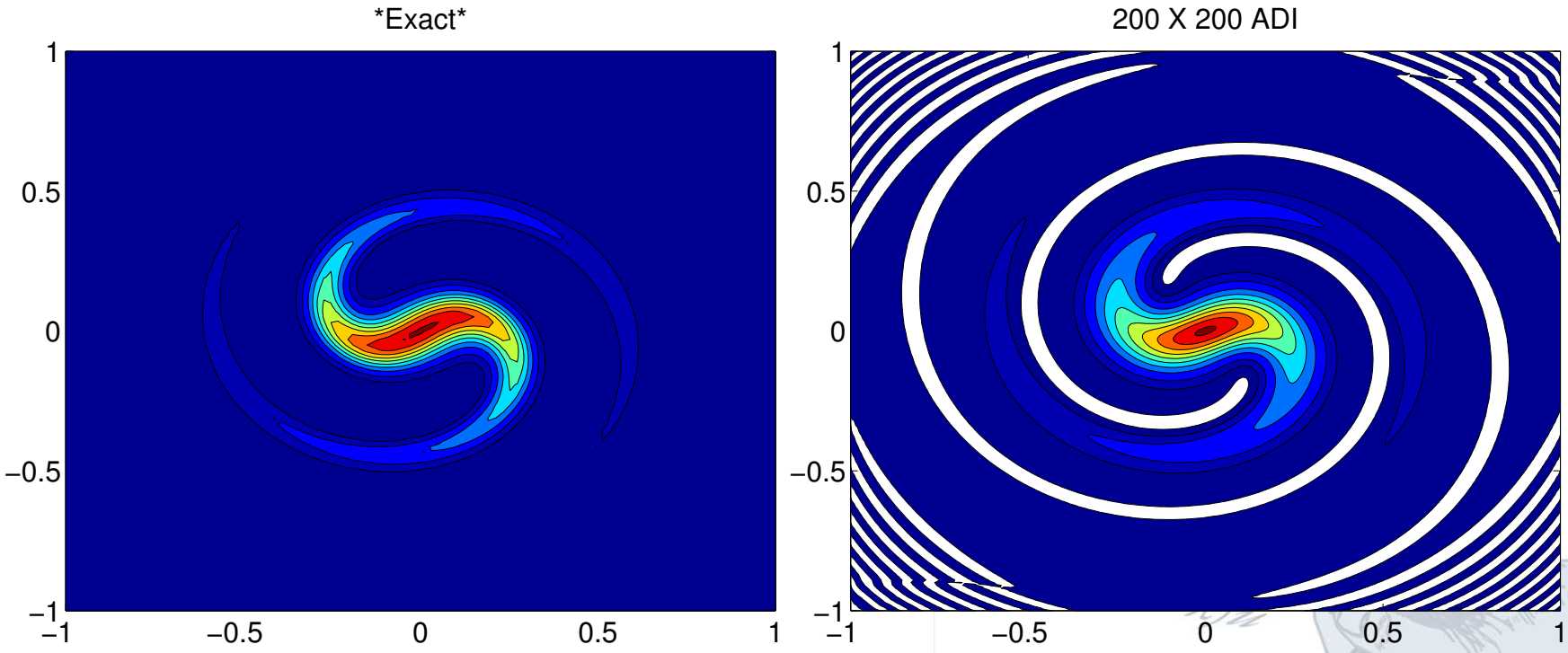

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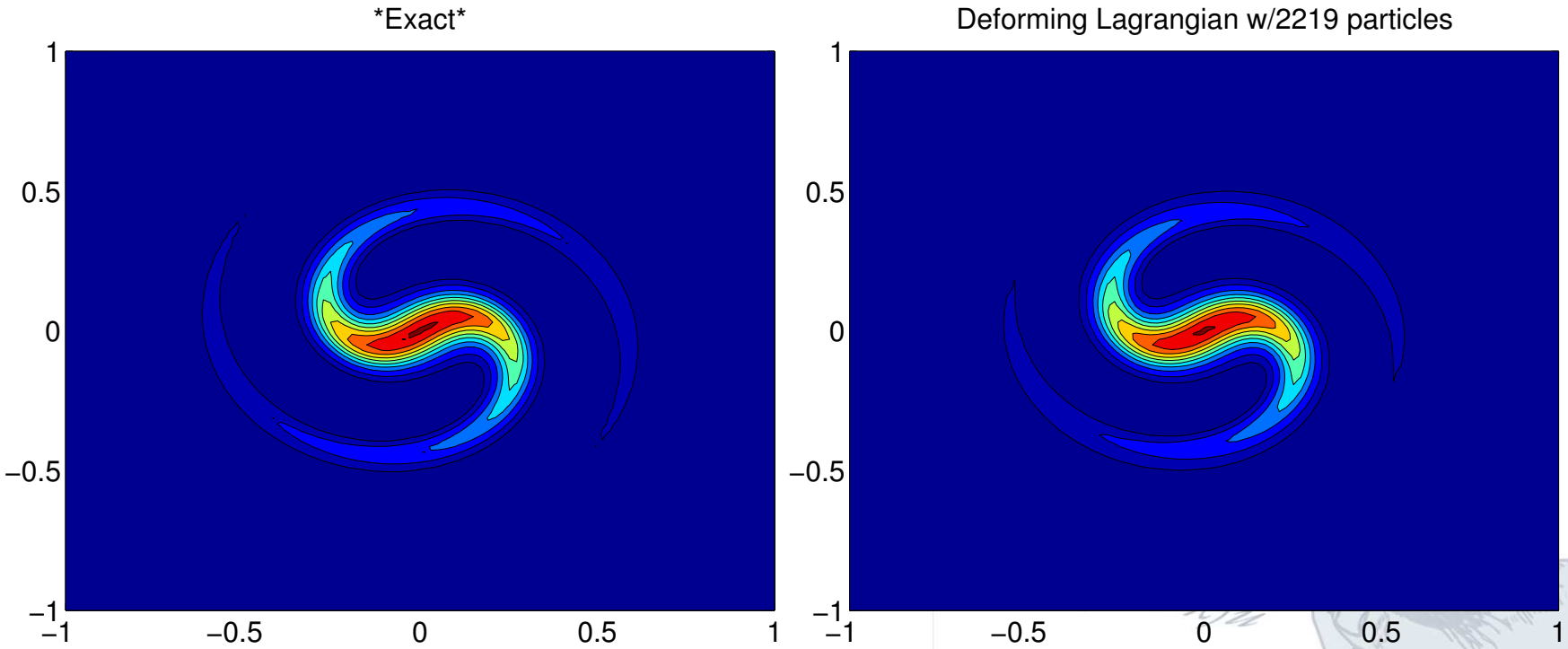

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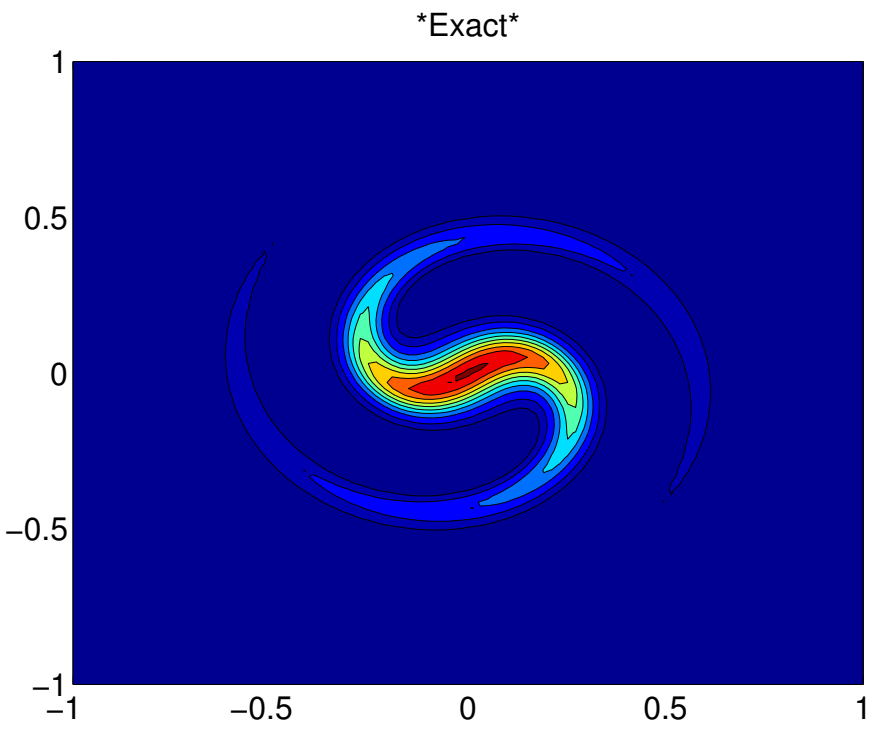
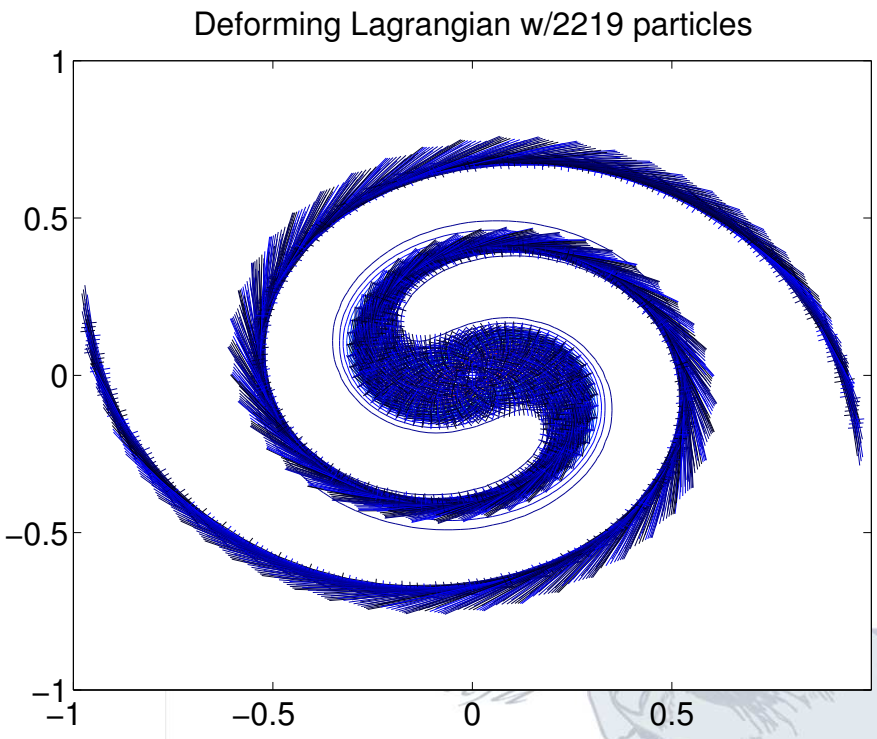




Collaborations

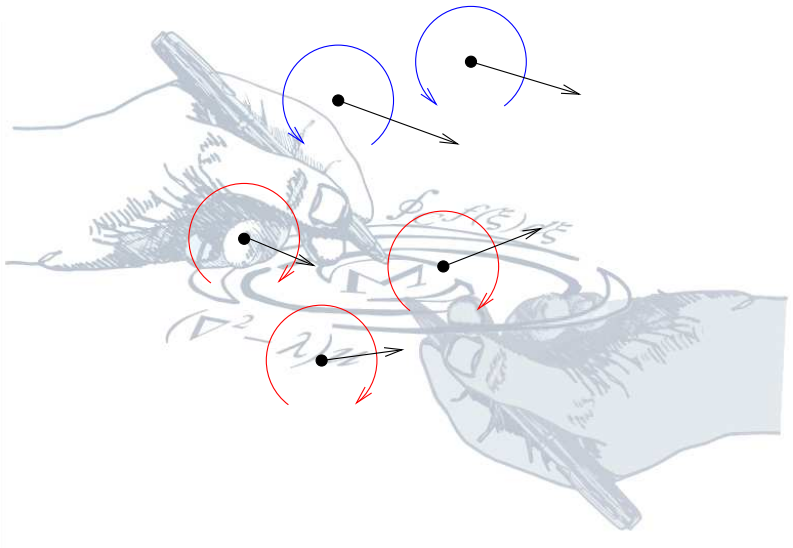
- Lorena Barba (Mechanical, Boston University) - Field interpolation problem.
- Rodrigo Platte (Mathematics, Arizona State/Oxford U) - Spectral interpolation and the Biot-Savart integral
- Stephen Siegel and Sam Moelius (Computer and information sciences, UD) - Parallel implementations
- Travis Mitchell (Physics, UD) - Linear and nonlinear evolution of electron vortices

Vortex methods

$$\frac{D\omega}{Dt} = \frac{1}{\mathrm{Re}}\nabla^2\omega$$

Express ω (or $\vec{\omega}$ in 3D) as a linear combination of moving basis functions (blobs):

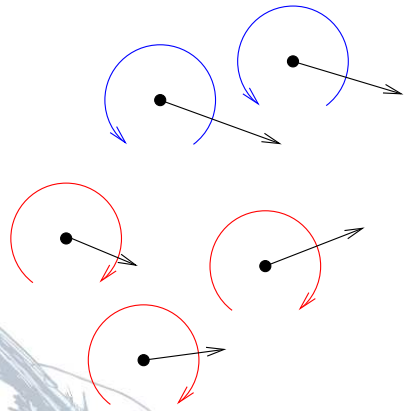
$$\begin{aligned}\widehat{\omega}(\vec{x}) &= \sum_{i=1}^{N}\gamma_i\phi(\vec{x}-\vec{x}_i)\\ \frac{d\vec{x}_i}{dt} &= \vec{u}(\vec{x}_i)\end{aligned}$$



The Biot-Savart integral

$$\widehat{\omega}(\vec{x}) = \sum_{i=1}^{N} \gamma_i \phi(\vec{x} - \vec{x}_i)$$

$$\frac{d\vec{x}_i}{dt} = \vec{u}(\vec{x}_i)$$

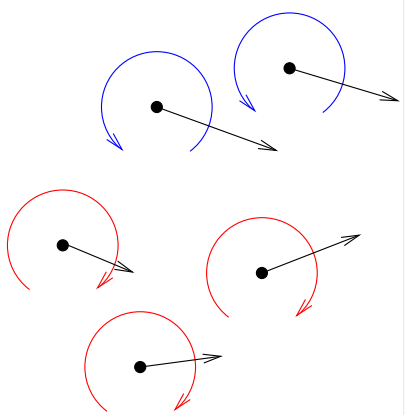


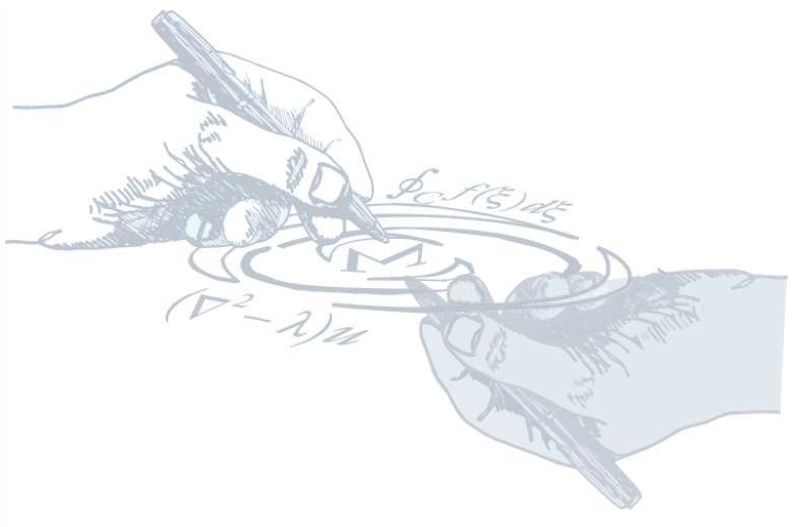
Since ω ($\widehat{\omega}$) is our primitive variable, we must determine $\vec{u}$ from a Biot-Savart integral via a streamfunction ψ. ($\nabla^2\psi = -\widehat{\omega}$)

An example with Gaussian blobs

$$\phi(\vec{x}) = \frac{1}{4\pi\sigma^2}\exp\left(-\frac{|\vec{x}|^2}{4\sigma^2}\right)$$

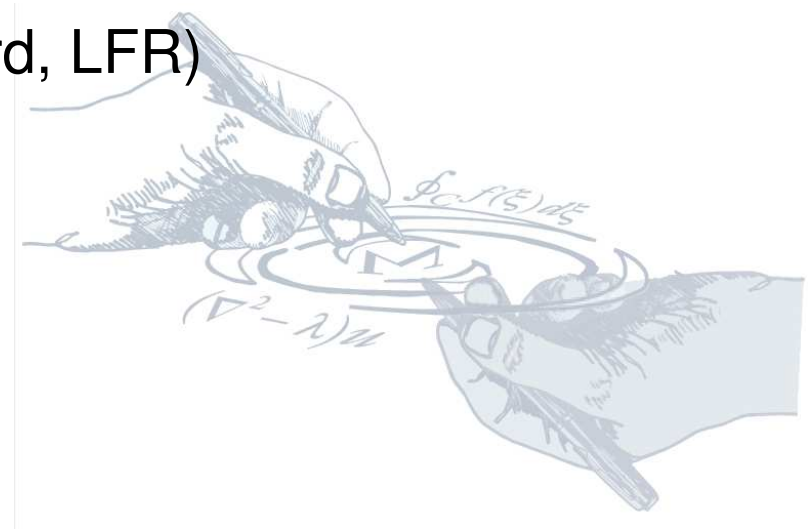
$$\vec{u}(\vec{x}) = \sum_{i=1}^{N}\frac{\gamma_i}{2\pi}\frac{\vec{x}-\vec{x}_i}{|\vec{x}-\vec{x}_i|^2}\left[1-\exp\left(-\frac{|\vec{x}-\vec{x}_i|^2}{4\sigma^2}\right)\right]$$





Approaches to viscous diffusion

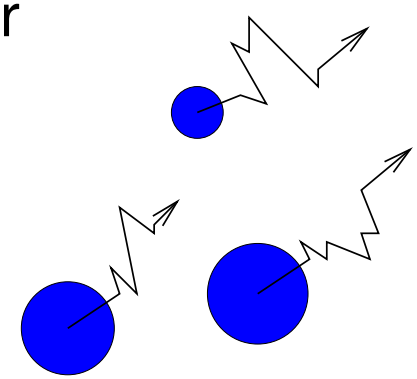
- Random walks (Chorin, Sethian, others)
- Circulation redistribution (Mas Gallic, Cottet, Winklemanns, others)
- Corrected core spreading (Leonard, LFR)



Approaches to viscous diffusion

- Random walks (Chorin, Sethian, others)
  Poor performance. Relies upon operator splitting.

$$\frac{D\omega}{Dt} = 0 \longleftrightarrow \frac{\partial \omega}{\partial t} = \frac{1}{\mathrm{Re}} \nabla^2 \omega$$



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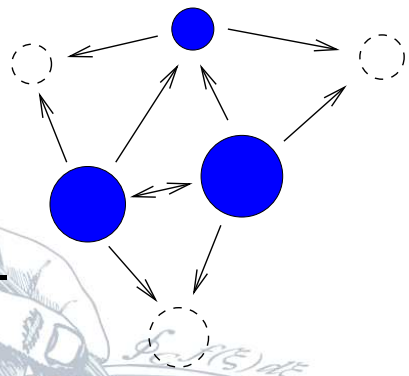

  Modest performance but not naturally adaptive.
- Corrected core spreading (Leonard, LFR)

Approaches to viscous diffusion

- Random walks (Chorin, Sethian, others)
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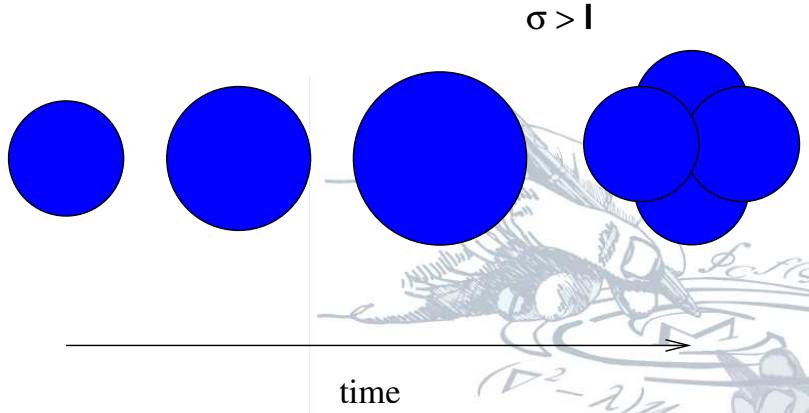
Good performance but requires field interpolation.



Corrected core spreading vortex method (CCVSM)

$$
\begin{aligned}
\phi(\vec{x}) &= \frac{1}{4\pi\sigma_i^2}\exp\left(-\frac{|\vec{x}|^2}{4\sigma_i^2}\right), \\
\vec{u}(\vec{x}) &= \sum_{i=1}^{N}\frac{\gamma_i}{2\pi}\frac{\vec{x}-\vec{x}_i}{|\vec{x}-\vec{x}_i|^2}\left[1-\exp\left(-\frac{|\vec{x}-\vec{x}_i|^2}{4\sigma_i^2}\right)\right], \\
\frac{d\sigma_i}{dt} &= \frac{1}{\mathrm{Re}}.
\end{aligned}
$$

Some comments: Field interpolation can maintain spatial accuracy for all time.

Systematic improvement of spatial accuracy

$$R\widehat{\omega} = \partial_t \widehat{\omega} + \vec{u} \cdot \nabla \widehat{\omega} - \frac{1}{\mathrm{Re}} \nabla^2 \widehat{\omega}$$

For the exact solution, $R\omega = 0$.

The residual $R\widehat{\omega}$ is a mathematical structure determining a set of **evolution equations** for the basis function parameters, and it provides an **upper bound on the spatial errors**.

Intuitive ideas.

Axisymmetric Gaussian basis functions achieve second order accuracy $O(\sigma^2)$ and exactly solve

$$\partial_t \phi + \vec{u}(\vec{x}_i) \cdot \nabla \phi - \frac{1}{\mathrm{Re}} \nabla^2 \phi = 0.$$

Deforming elliptical Gaussian basis functions might achieve fourth order accuracy $O(\sigma^4)$ and solve

$$\partial_t \phi + [\vec{u}(\vec{x}_i) + D\vec{u}(\vec{x}_i)(\vec{x} - \vec{x}_i)] \cdot \nabla \phi - \frac{1}{\mathrm{Re}} \nabla^2 \phi = 0.$$

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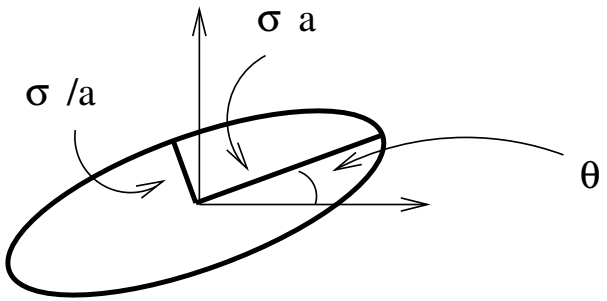
But this is not the full residual...

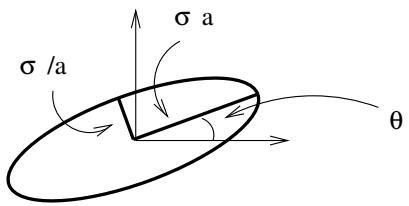
The dynamics of elliptical Gaussians



Find the evolution equations for $\vec{x}_i$, θ_i, σ_i and a_i so that $\phi \equiv \phi(\vec{x}; \vec{x}_i, \sigma_i, a_i, \theta_i)$ satisfies

$$\partial_t \phi + [\vec{u}(\vec{x}_i) + D\vec{u}(\vec{x}_i)(\vec{x} - \vec{x}_i)] \cdot \nabla \phi - \frac{1}{\mathrm{Re}} \nabla^2 \phi = 0.$$

The dynamics of elliptical Gaussians



For simplicity $\theta_i = 0$...

$$\frac{d}{dt}\vec{x}_i = \begin{bmatrix} u_i \\ v_i \end{bmatrix}$$

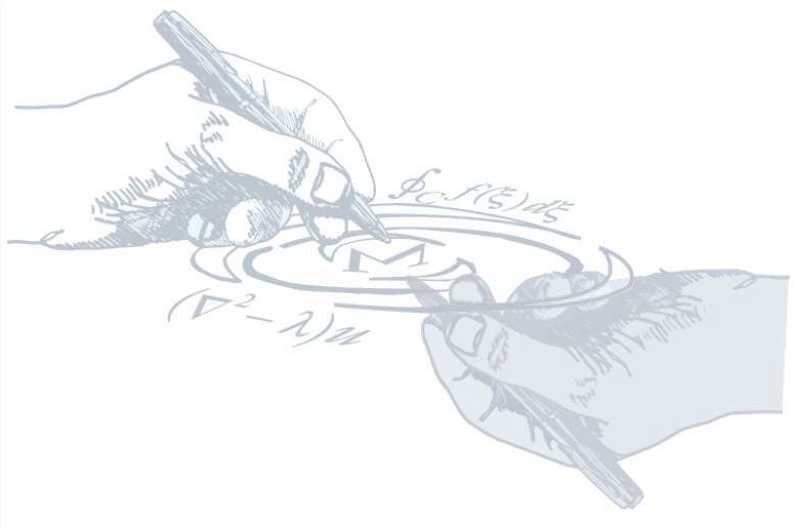
$$\frac{d}{dt}(\sigma_i^2) = \frac{1}{2\mathrm{Re}}(a_i^2 + a_i^{-2})$$

$$\frac{d}{dt}(a_i^2) = 2d_{11,i}a_i^2 + \frac{1}{2\mathrm{Re}\sigma_i^2}(1 - a_i^4)$$

$$\frac{d}{dt}\theta_i = \frac{d_{21,i} - d_{12,i}}{2} - \frac{d_{21,i} + d_{12,i}}{2}\frac{(a_i^{-2} + a_i^2)}{(a_i^{-2} - a_i^2)}$$

Curvature corrections

Deforming elements are not enough to achieve fourth order spatial accuracy.



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\widehat{\omega} &= \sum_{i=0}^{N} \gamma_i \phi(\vec{x}; \vec{x}_i, \sigma_i, a_i, \theta_i) \\
&\approx \iint \frac{\gamma(\vec{t})}{dA} \phi\left[\vec{x}; \vec{t}, \sigma(\vec{t}), a(\vec{t}), \theta(\vec{t})\right] dA \\
&\approx \iint \tilde{\omega}(\vec{t}) \phi\left[\vec{x}; \vec{t}, \sigma(\vec{t}), a(\vec{t}), \theta(\vec{t})\right] dA \\
&\approx \iint \tilde{\omega}(\vec{x} - \vec{t}) \phi\left[\vec{0}; \vec{t}, \sigma(\vec{t}), a(\vec{t}), \theta(\vec{t})\right] dA
\end{aligned}$$

$$R\widehat{\omega} = \partial_t \widehat{\omega} + \vec{u} \cdot \nabla \widehat{\omega} - \frac{1}{\mathrm{Re}} \nabla^2 \widehat{\omega}$$

Curvature corrections

Deforming elements are not enough to achieve fourth order spatial accuracy.

$$R\widehat{\omega} = \partial_t \widehat{\omega} + \vec{u} \cdot \nabla \widehat{\omega} - \frac{1}{\mathrm{Re}} \nabla^2 \widehat{\omega}$$

$$\widehat{\omega} \approx \iint \tilde{\omega}(\vec{x} - \vec{t}) \phi \left[\vec{0}; \vec{t}, \sigma(\vec{t}), a(\vec{t}), \theta(\vec{t}) \right] dA$$

$$\begin{aligned} R\widehat{\omega} &\approx \iint_{-\infty}^{\infty} \phi(\vec{t}) \nabla_{\vec{t}} \tilde{\omega}(\vec{x} - \vec{t}) \cdot \left\{ \frac{1}{2} D^2 \vec{u}(\vec{x})[\vec{t}, \vec{t}] + \ldots \right\} d\vec{t}, \\ &\approx \iint_{-\infty}^{\infty} \phi(\vec{t}) \nabla \tilde{\omega}(\vec{x}) \cdot \left\{ \frac{1}{2} D^2 \vec{u}(\vec{x})[\vec{t}, \vec{t}] + \ldots \right\} d\vec{t}, \end{aligned}$$

which is no better than a non-deforming method because the second moment of an elliptical Gaussian will be $O(l^2)$.

Curvature corrections (cont'd)

$$R\widehat{\omega} \approx \iint_{-\infty}^{\infty} \phi(\vec{t})\nabla\tilde{\omega}(\vec{x}) \cdot \left\{ \frac{1}{2} D^2 \vec{u}(\vec{x})[\vec{t},\vec{t}] + \ldots \right\} d\vec{t}$$

To erase the leading order, we choose a $\widehat{u}$ different from $\vec{u}$.

$$\iint_{-\infty}^{\infty} \phi(\vec{t}) D^2 u(\vec{x})[\vec{t},\vec{t}] d\vec{t} = 2\sigma^2 \left(u_{xx} M_{xx} + 2u_{xy} M_{xy} + u_{yy} M_{yy} \right),$$

where

$$\begin{aligned} M_{xx} &= \cos^2\theta a^2 + \sin^2\theta / a^2, \\ M_{xy} &= \cos\theta \sin\theta (a^2 - a^{-2}), \\ M_{yy} &= \cos^2\theta / a^2 + \sin^2\theta a^2. \end{aligned}$$

Curvature corrections (cont'd)

$$R\widehat{\omega} \approx \iint_{-\infty}^{\infty} \phi(\vec{t}) \nabla \tilde{\omega}(\vec{x}) \cdot \left\{ \frac{1}{2} D^2 \vec{u}(\vec{x})[\vec{t}, \vec{t}] + \ldots \right\} d\vec{t}$$

To erase the leading order, we choose a $\widehat{u}$ different from $\vec{u}$. Thus, we can eliminate the leading order error, by choosing

$$\widehat{u}_i = \vec{u}(\vec{x}_i) + \sigma^2 \left(u_{xx} M_{xx} + 2 u_{xy} M_{xy} + u_{yy} M_{yy} \right)$$

The next contribution to the error is $O(l^4)$.

The dynamics of elliptical Gaussians

$$\frac{d}{dt}\vec{x}_i = \begin{bmatrix} u_i \\ v_i \end{bmatrix} + 2\sigma_i^2 \begin{bmatrix} u_{xx,i}M_{xx} + 2u_{xy,i}M_{xy} + u_{yy,i}M_{yy} \\ v_{xx,i}M_{xx} + 2v_{xy,i}M_{xy} + v_{yy,i}M_{yy} \end{bmatrix}$$

$$\frac{d}{dt}(\sigma_i^2) = \frac{1}{2\mathrm{Re}}(a_i^2 + a_i^{-2})$$

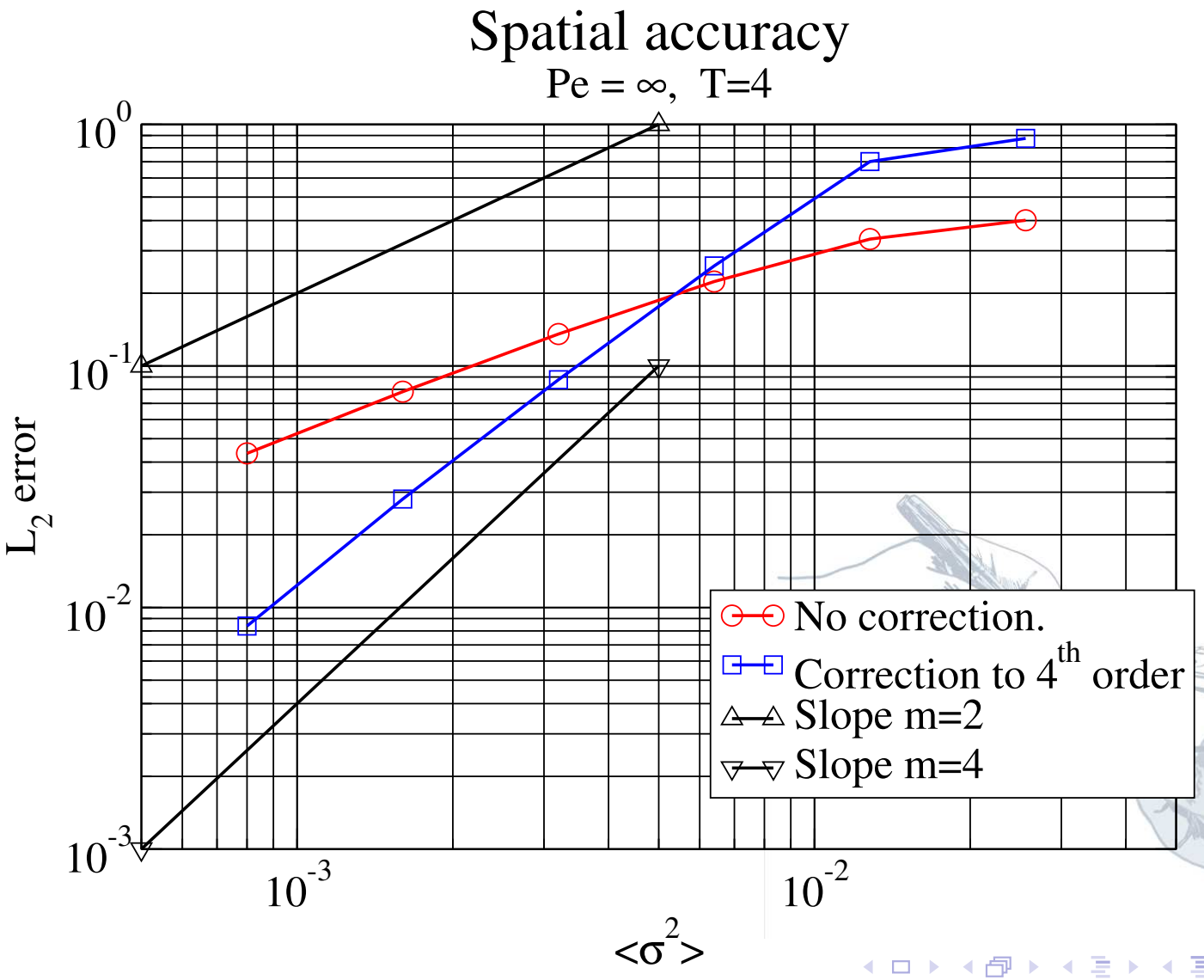
$$\frac{d}{dt}(a_i^2) = 2d_{11,i}a_i^2 + \frac{1}{2\mathrm{Re}\sigma_i^2}(1 - a_i^4)$$

$$\frac{d}{dt}\theta_i = \frac{d_{21,i} - d_{12,i}}{2} - \frac{d_{21,i} + d_{12,i}}{2}\frac{(a_i^{-2} + a_i^2)}{(a_i^{-2} - a_i^2)}$$

where

$$M_{xx} = c^2a^2 + s^2/a^2, \quad M_{xy} = cs(a^2 - a^{-2}), \quad M_{yy} = c^2/a^2 + s^2a^2.$$

Convergence properties



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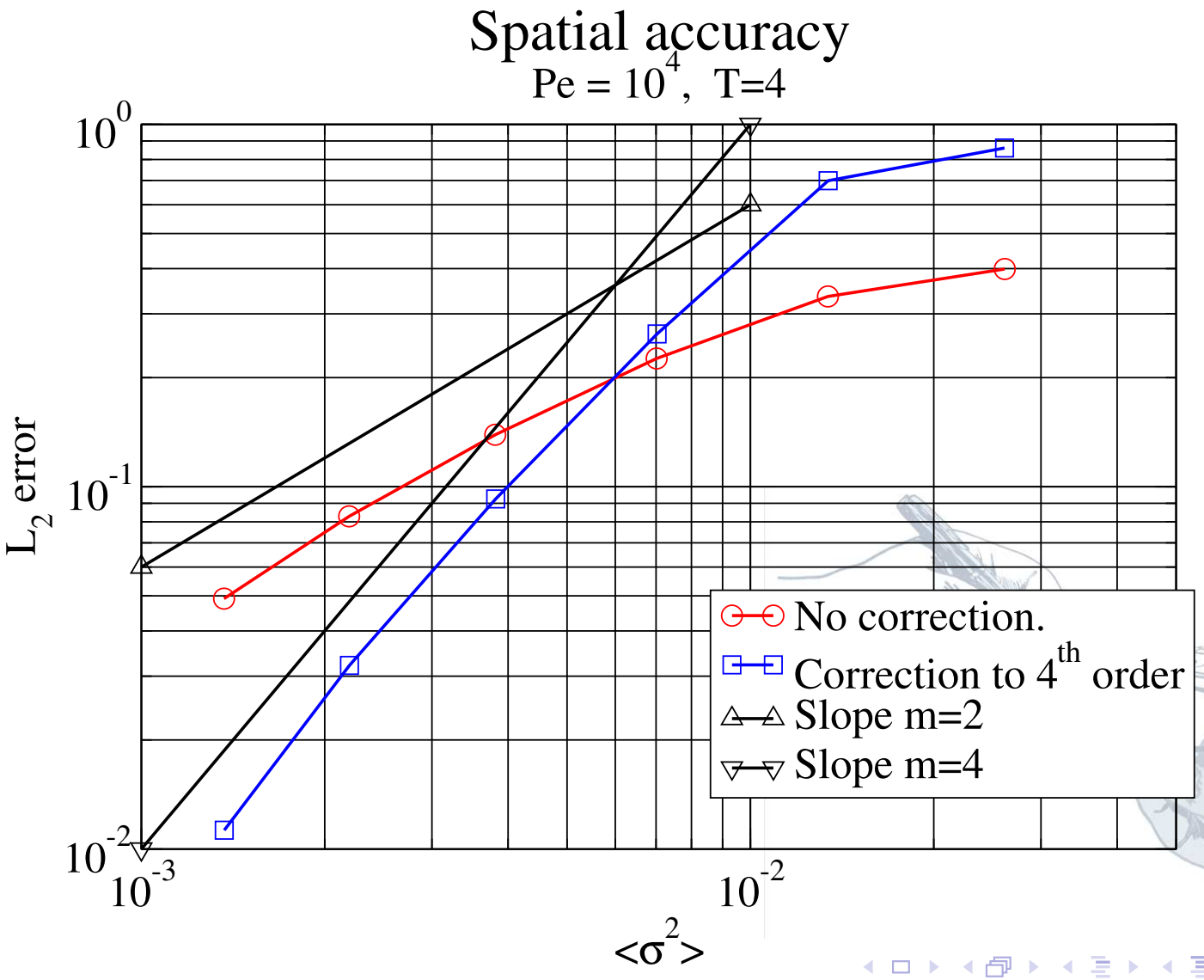

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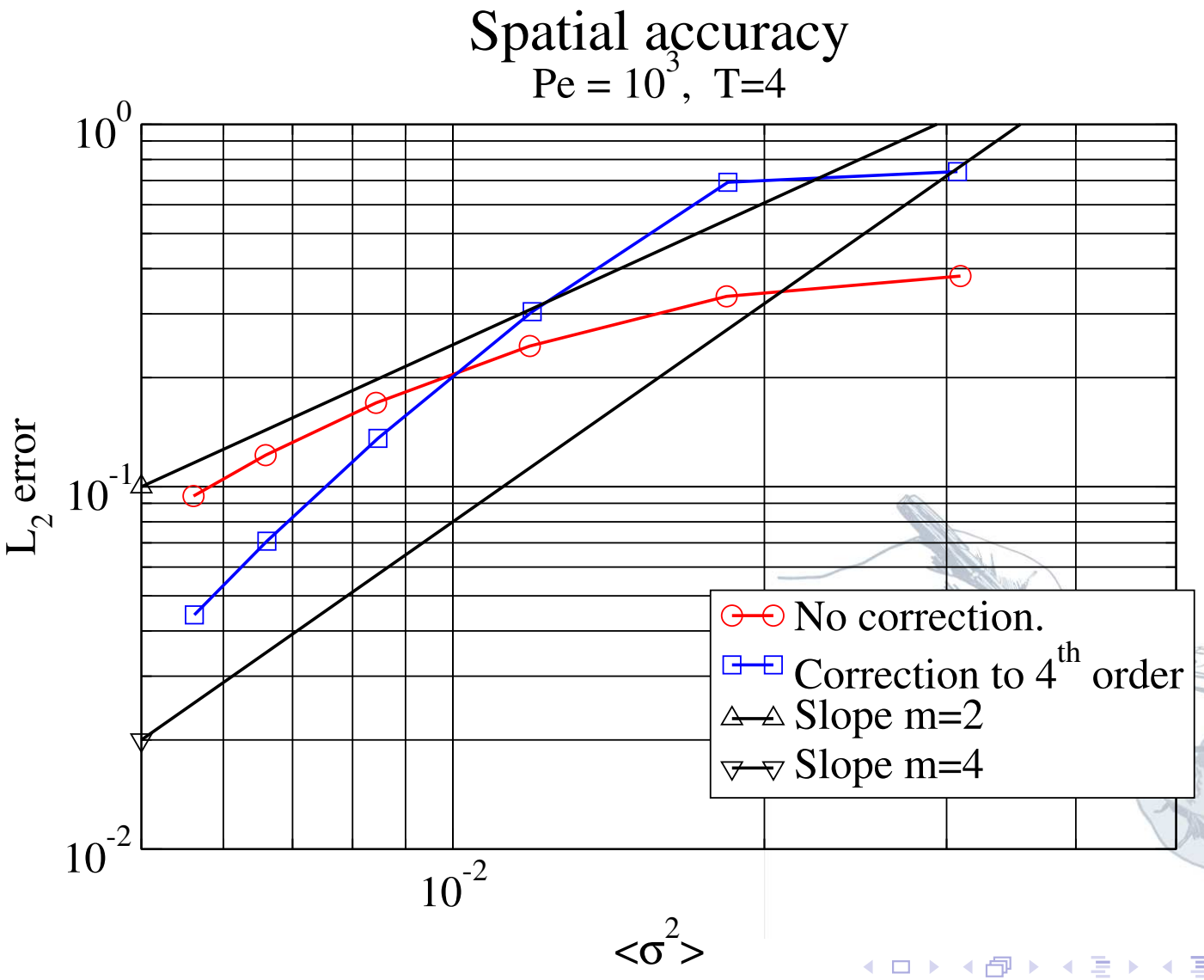
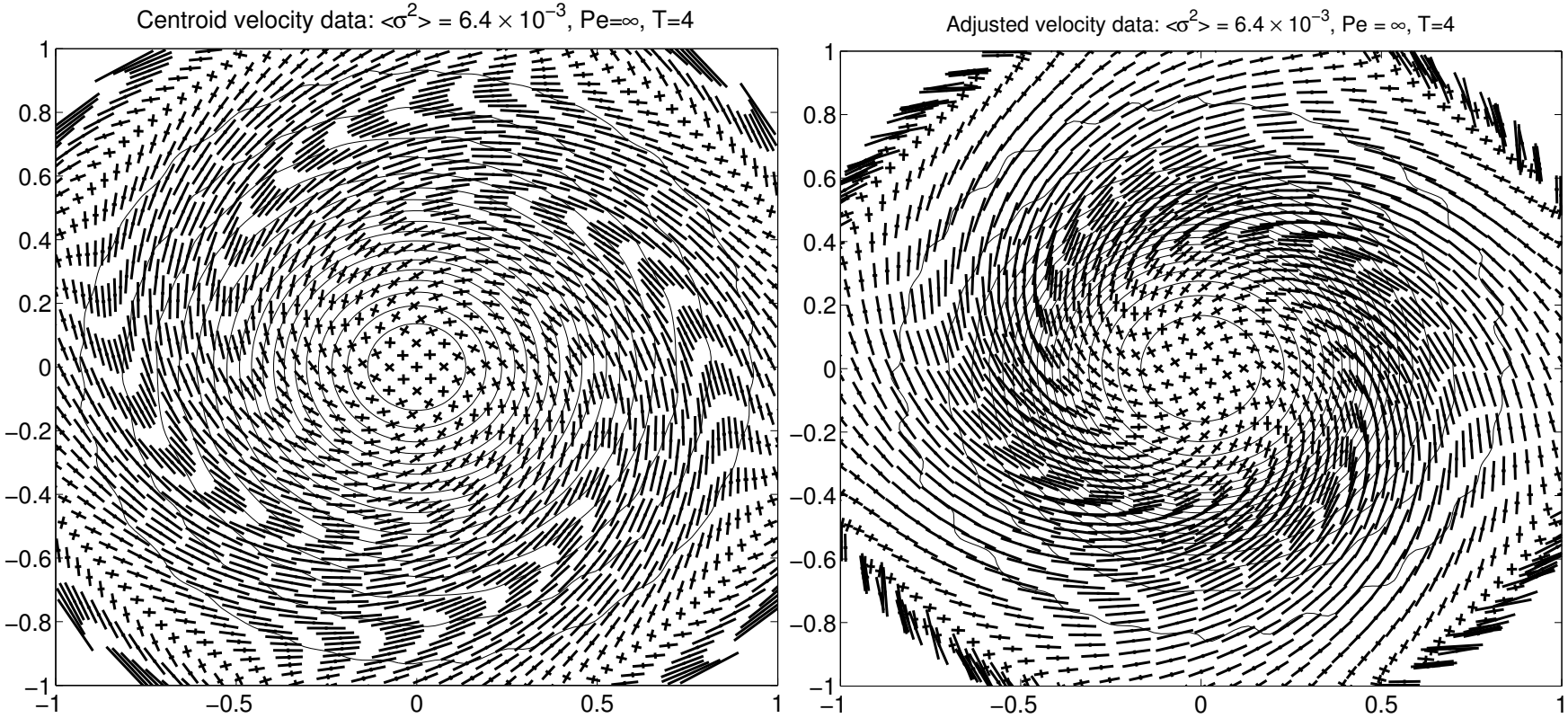


Unconventional wisdom

Lagrangian elements do not necessarily need to follow physical streamlines at $\mathrm{Pe} = 10^4$.



Field interpolation problem

Given a function ω_1, determine γ's on a distribution $(\vec{y})$

$$\omega_2 = \sum_{i=1}^{N} \gamma_i' \phi(\vec{y} - \vec{y}_i; \sigma_0, 1, 0)$$

so that $\|\omega_1 - \omega_2\|$ is as small as possible.
You need not assume that the $\vec{y}$ is an isotropic distribution with a mesh spacing of h.

1. We do this to project initial conditions onto a set of particles.
2. We do this to replace an inaccurate configuration with an accurate configuration.

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Methods of solution

1. Ignore the problem and use $\gamma' = \omega_1 h^2$. Con: Large errors.
2. Reinterpolate the γ's onto new positions. $\gamma \to \gamma'$. Cons: Assumes you know an old γ to reinterpolate. It only works for isotropic elements.
3. Use RBF interpolation with a fixed point iteration ($\omega_1 \to \gamma'$). Con: It will not converge if h/σ is small.
4. Use RBF interpolation ($\omega_1 \to \gamma'$). Cons: Ill-conditioned if h/σ is small or N is large. Slow convergence using standard solvers.
5. Use backward heat equation preconditioned RBF interpolation ($\omega_1 \to \gamma'$). This is essentially using method 1 and then improving it.

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RBF approach

Solve

$$\sum_{i=1}^{N} \gamma_i' \phi(\vec{y}_i; \sigma_0, 1, 0) = \omega_1$$

at the $\vec{y}$ collocation points.

This reduces to

$$A\vec{\gamma}' = \vec{\omega}_1$$

where A is ill-conditioned.

Reverse heat equation

Approach: Using $\gamma' = \omega h^2$, is an exact representation for a regularized version of ω_1.

$$\omega_2 = \sum_{i=1}^{N} \gamma_i' \phi(\vec{y}_i; \sigma_0, 1, 0)$$
$$\approx \iint \frac{\omega(\vec{y})}{4\pi\sigma^2} \exp\left(-\frac{|\vec{x} - \vec{y}|^2}{4\sigma^2}\right) d\vec{y}$$

If we choose $\gamma' = \omega d\vec{y} = \omega_1(\vec{y}_i) h^2$, we'll get the wrong answer.

Reverse heat equation

Since,

$$\omega_2 = \sum_{i=1}^{N} \gamma_i' \phi(\vec{y}_i; \sigma_0, 1, 0) \approx \iint \frac{\omega(\vec{y})}{4\pi\sigma^2} \exp\left(-\frac{|\vec{x}-\vec{y}|^2}{4\sigma^2}\right) d\vec{y}$$

we need to reverse the smoothing.
We would like to solve the problem

$$\partial_t \omega = \nabla^2 \omega, \quad \omega(\vec{x}, \sigma^2) = \omega_1(\vec{x})$$

and find the solution at $t = 0$.

This problem surfaced in image processing and deblurring long ago.

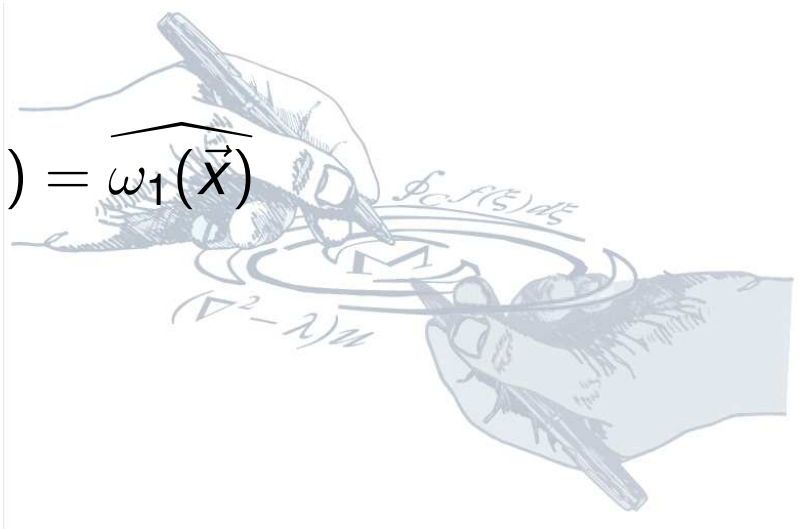
The reverse heat equation is ill-posed.

Fourier domain:

$$\omega = \int \widehat{\omega}(\vec{k}, t) e^{i\vec{k}\vec{x}} d\vec{k},$$

High frequencies (round-off and numerical errors) grow exponentially large.

$$\frac{\partial \widehat{\omega}}{\partial t} = -|\vec{k}|^2 \widehat{\omega}, \quad \widehat{\omega}(\vec{k}, \sigma^2) = \widehat{\omega_1(\vec{x})}$$



Solving the RHE

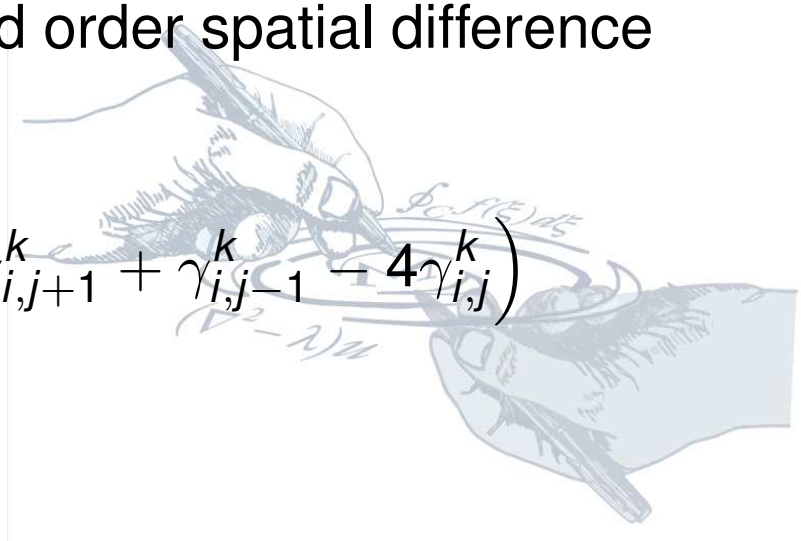
Solve the RHE with one temporal step to avoid stability issues.

- Are the errors dominated by spatial or temporal truncations?
- What is the role of discretization (overlap ratio) error $\beta = h^2/\sigma^2 = h^2/k$?

Example: Euler's method with a second order spatial difference (Δ_2):

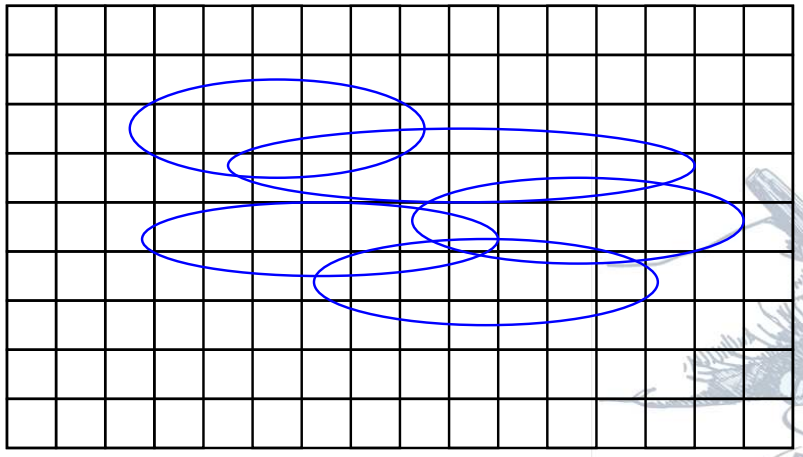
$$\gamma_{i,j}^{k+1} = \gamma_{i,j}^{k} + \frac{1}{\beta}\left(\gamma_{i+1,j}^{k} + \gamma_{i-1,j}^{k} + \gamma_{i,j+1}^{k} + \gamma_{i,j-1}^{k} - 4\gamma_{i,j}^{k}\right)$$

But, we'll just take one step.



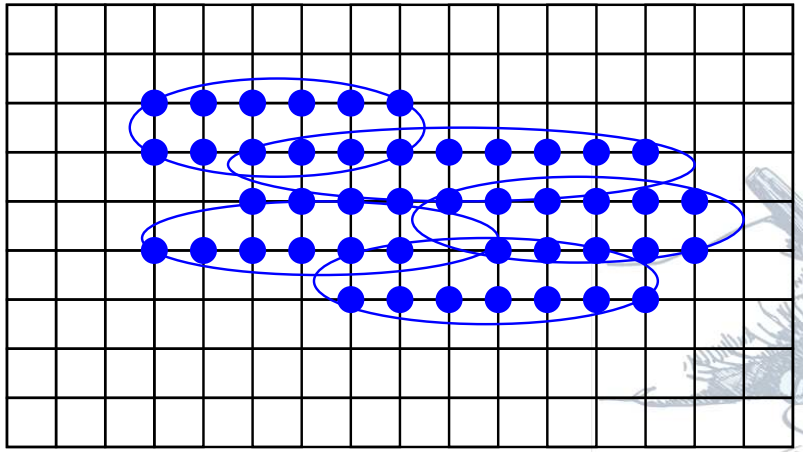
The algorithm

Step 1: Put a grid on an area of interest.



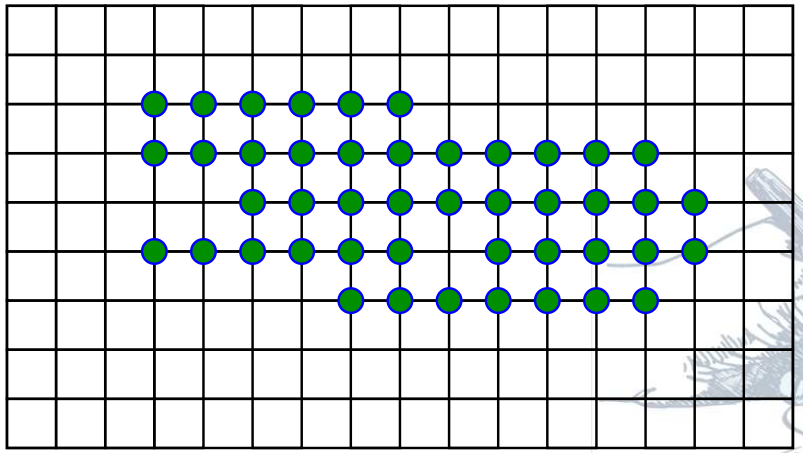
The algorithm

Step 2: Interpolate ωh^2 onto the grid.



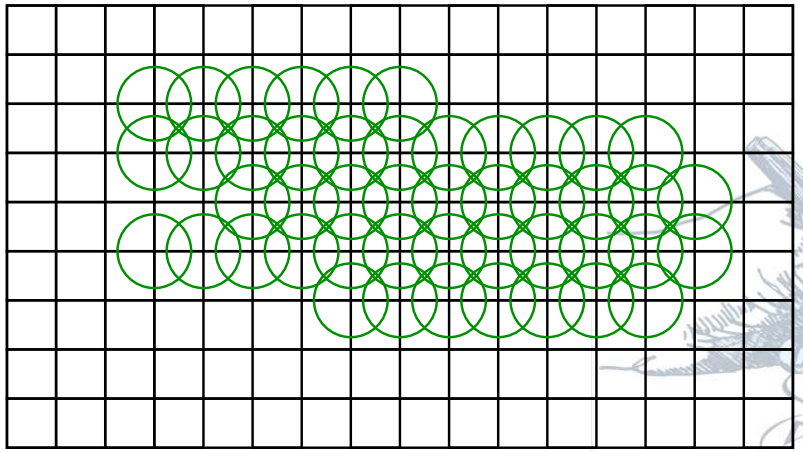
The algorithm

Step 3: Take one step of the RHE on the grid.

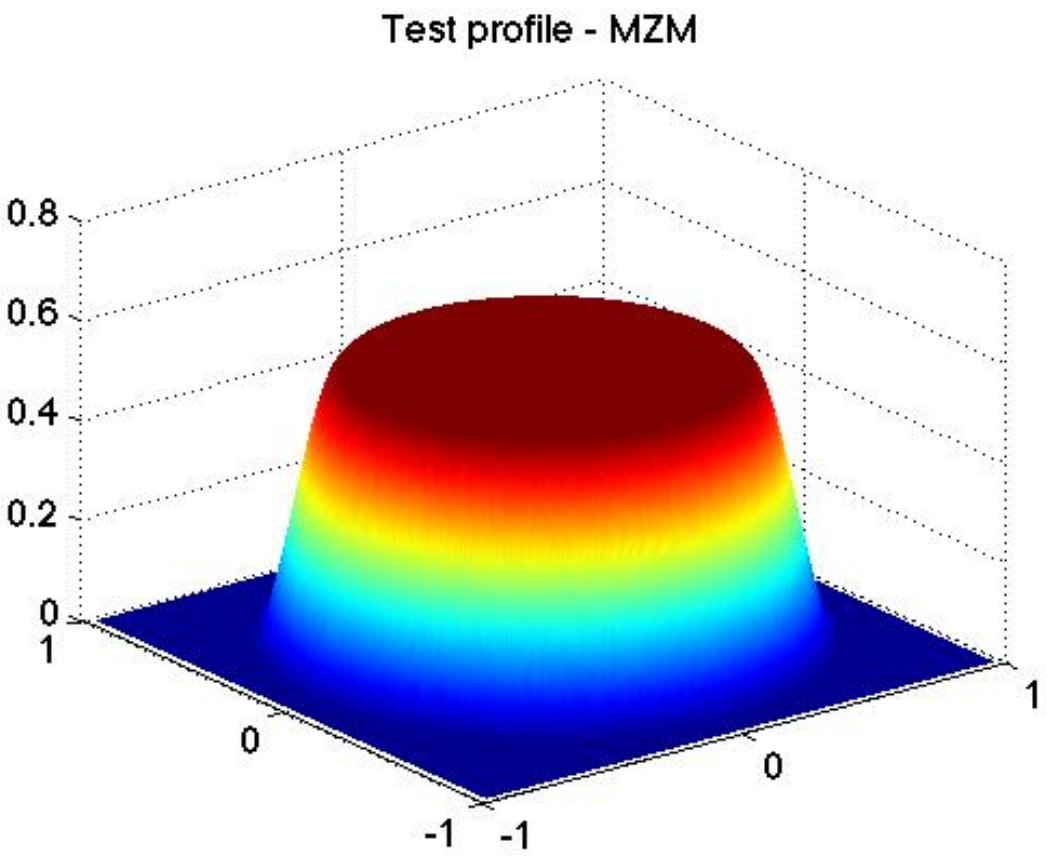


The algorithm

Step 4: Use the solution for γ. (Optional but important: Throw away particles with very weak circulations.)

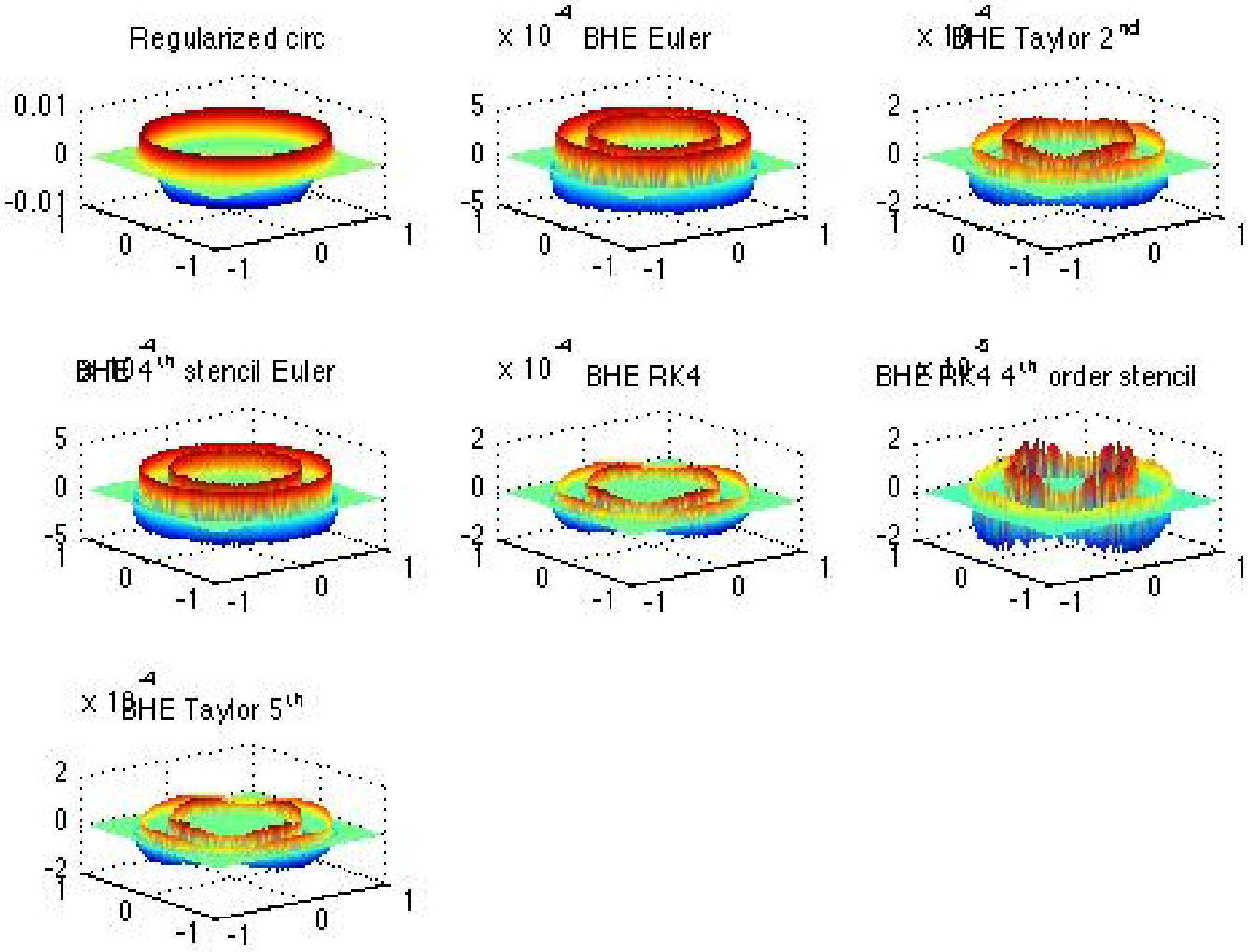


An example

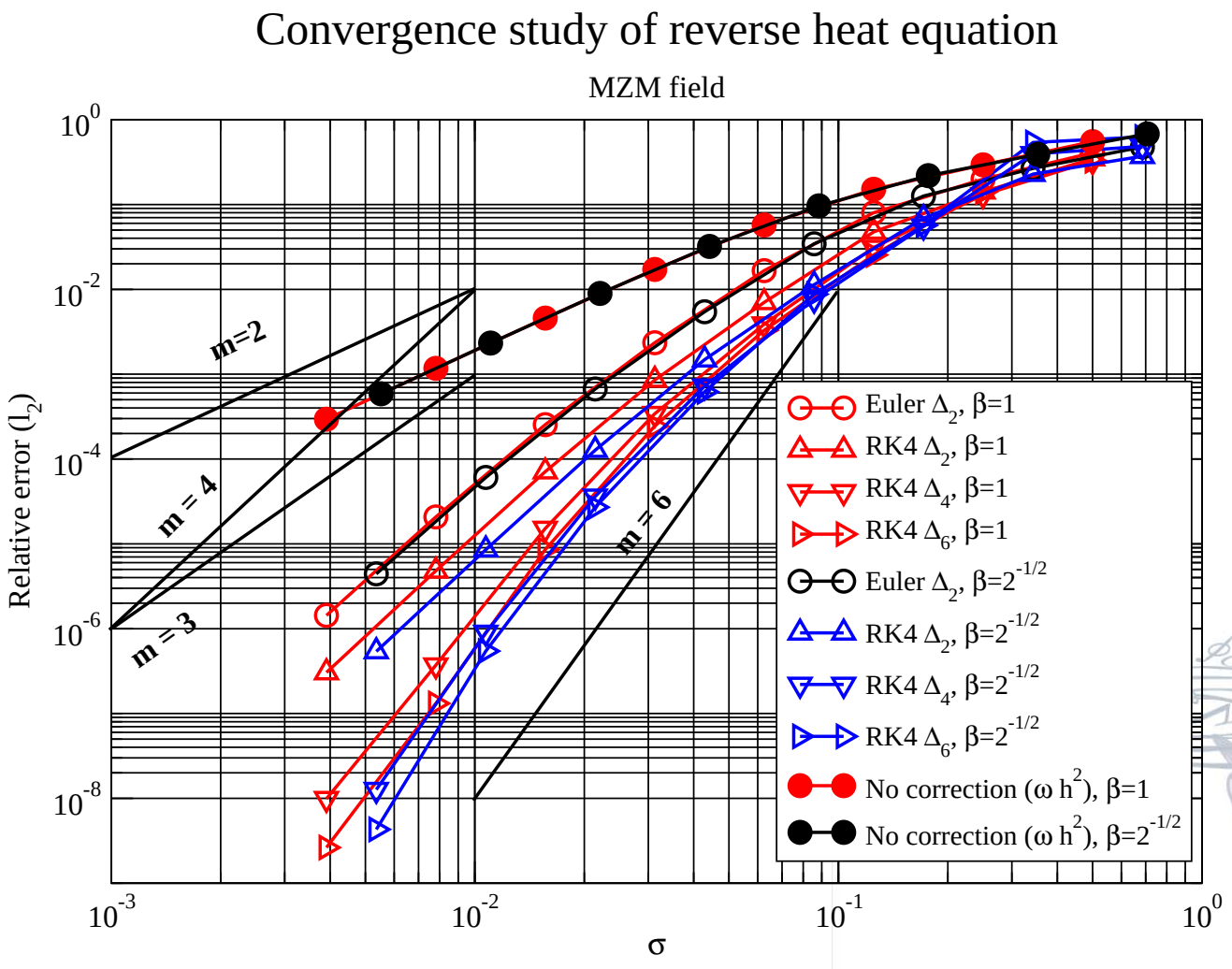


A patch with a sharp C^∞ transition.

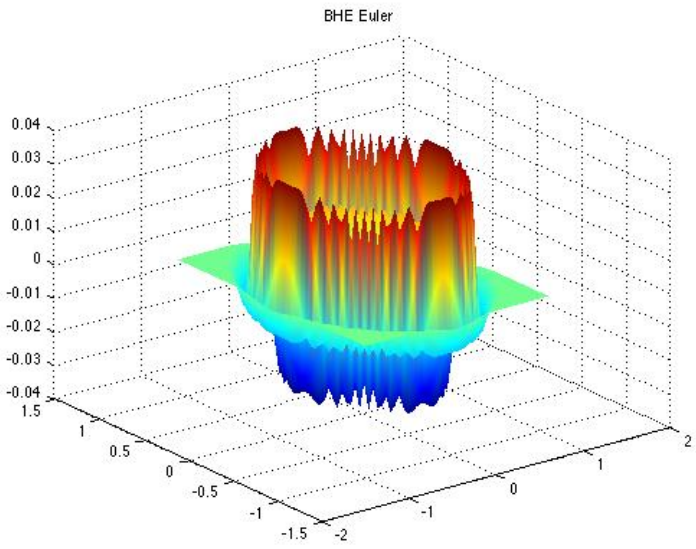
Solving the reverse heat equation ($\sigma = h = 10^{-2}$).

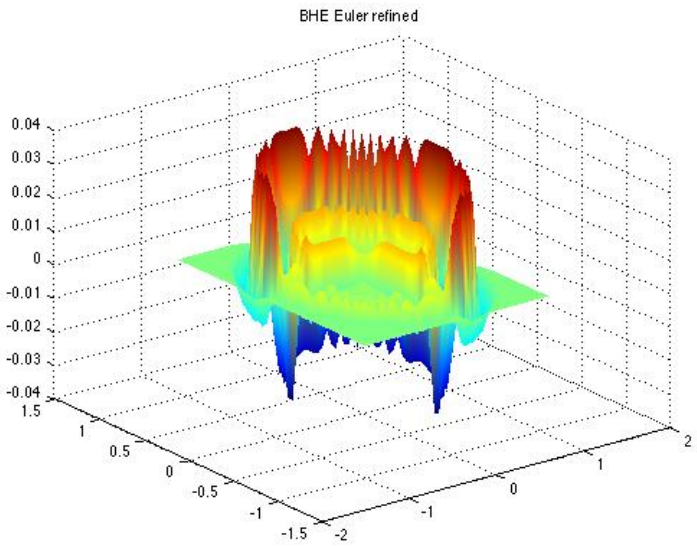


Convergence rates



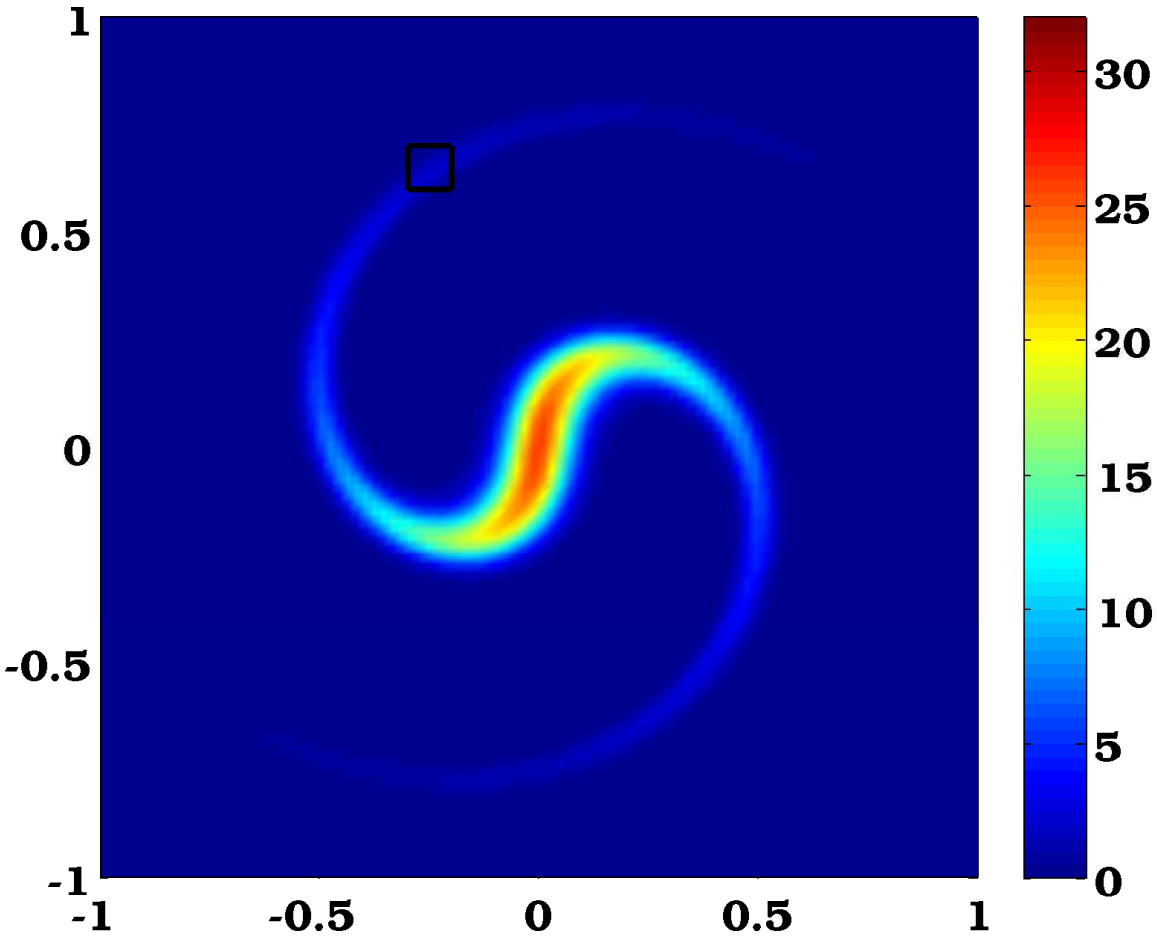
Adaptive reverse heat equation





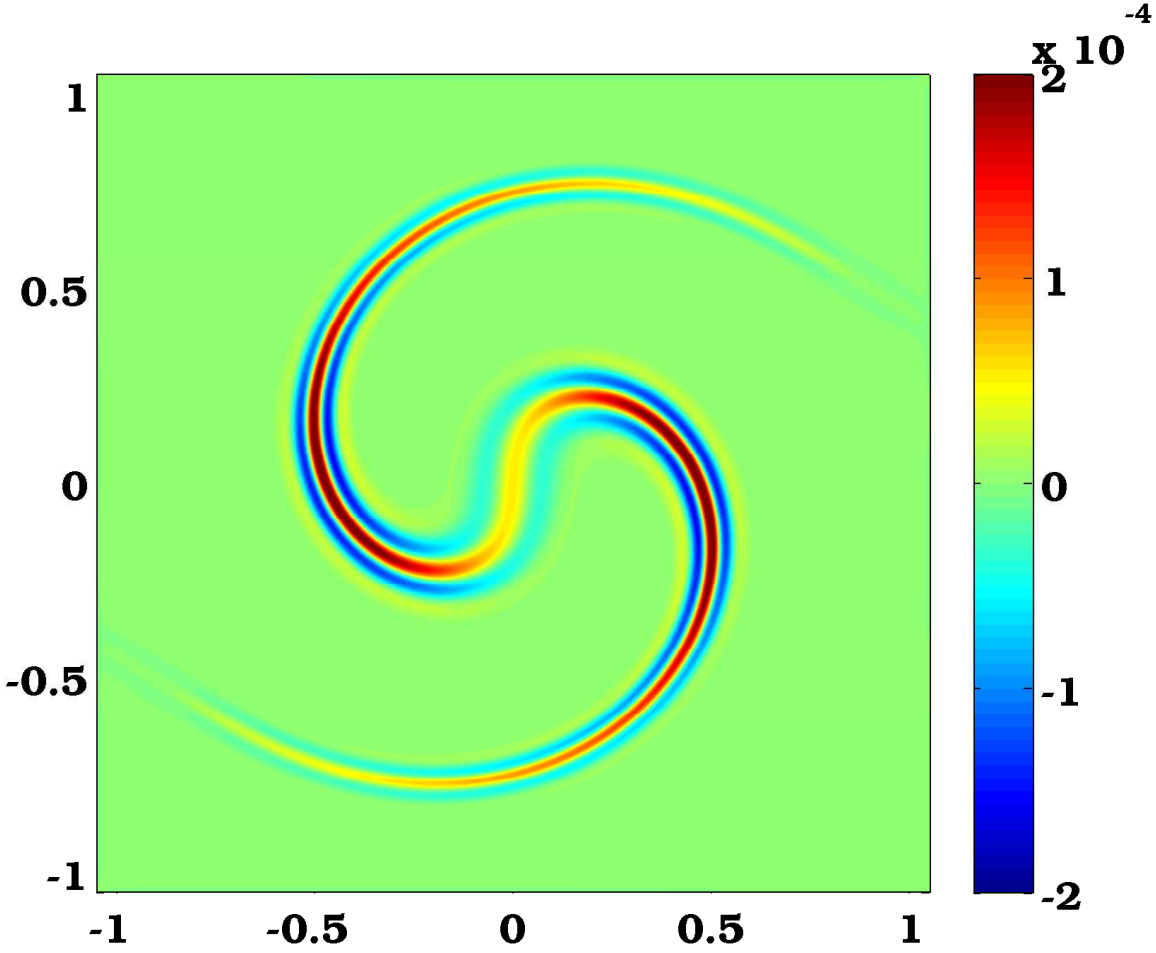
Verification of spatial errors

A test pattern high spatial frequencies.



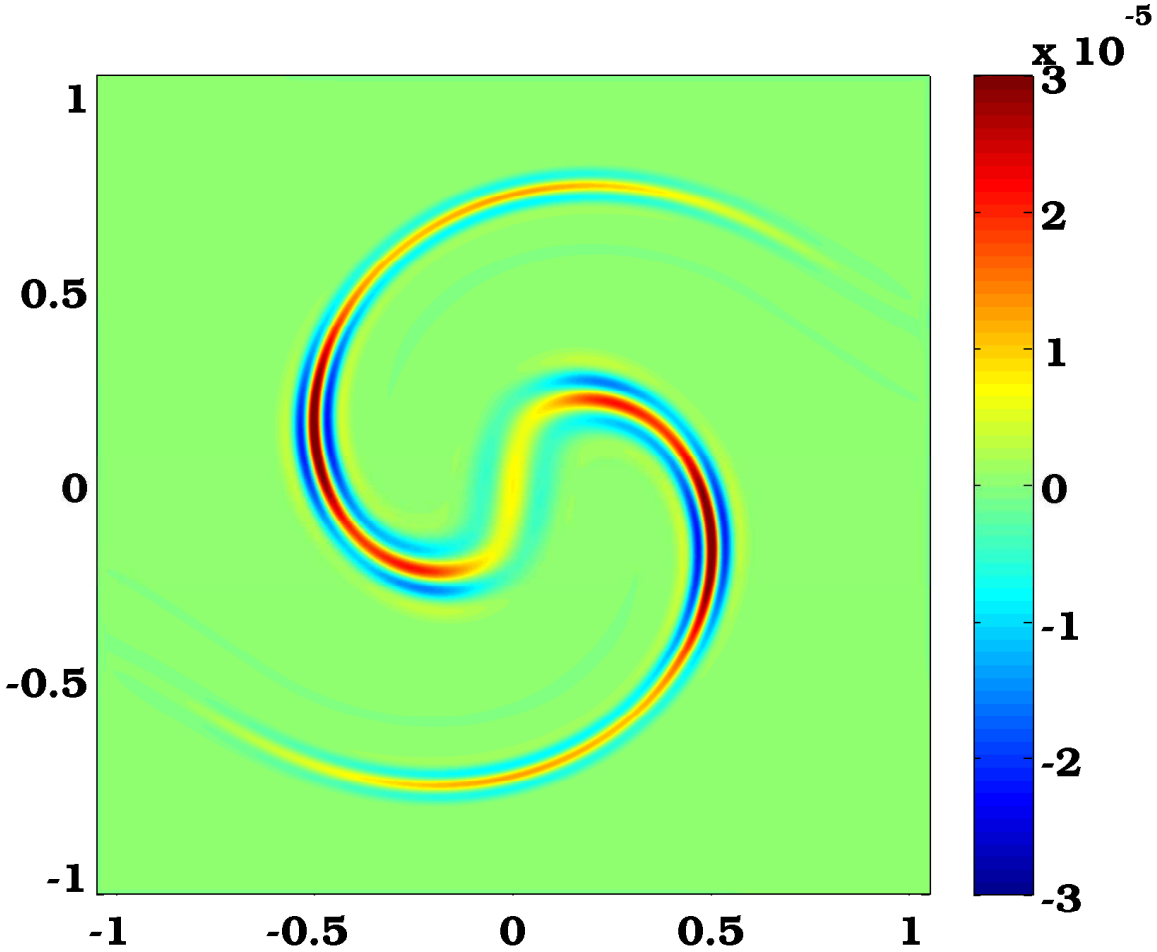
Verification of spatial errors

Euler's method in time. First order stencil.



Verification of spatial errors

RK4 in time. First order stencil.



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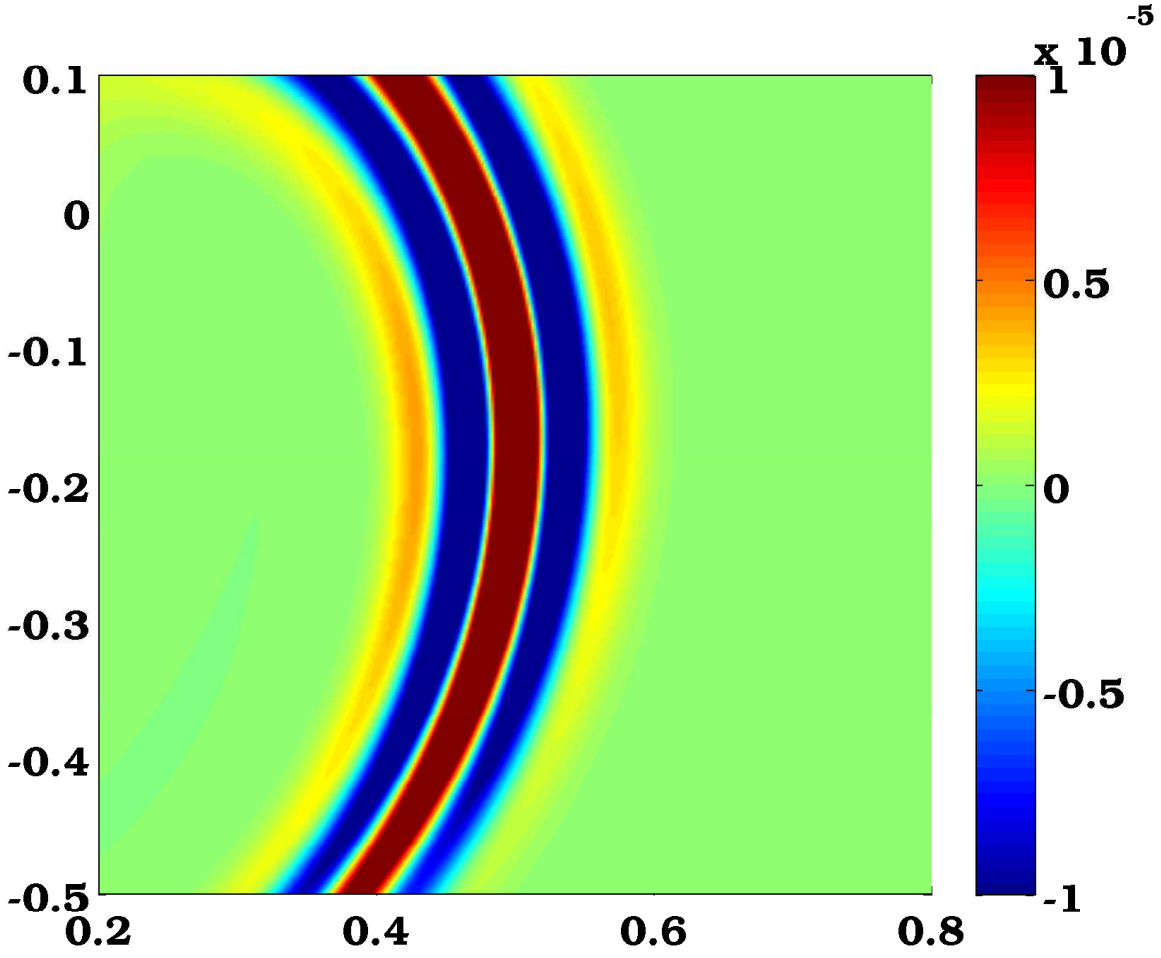
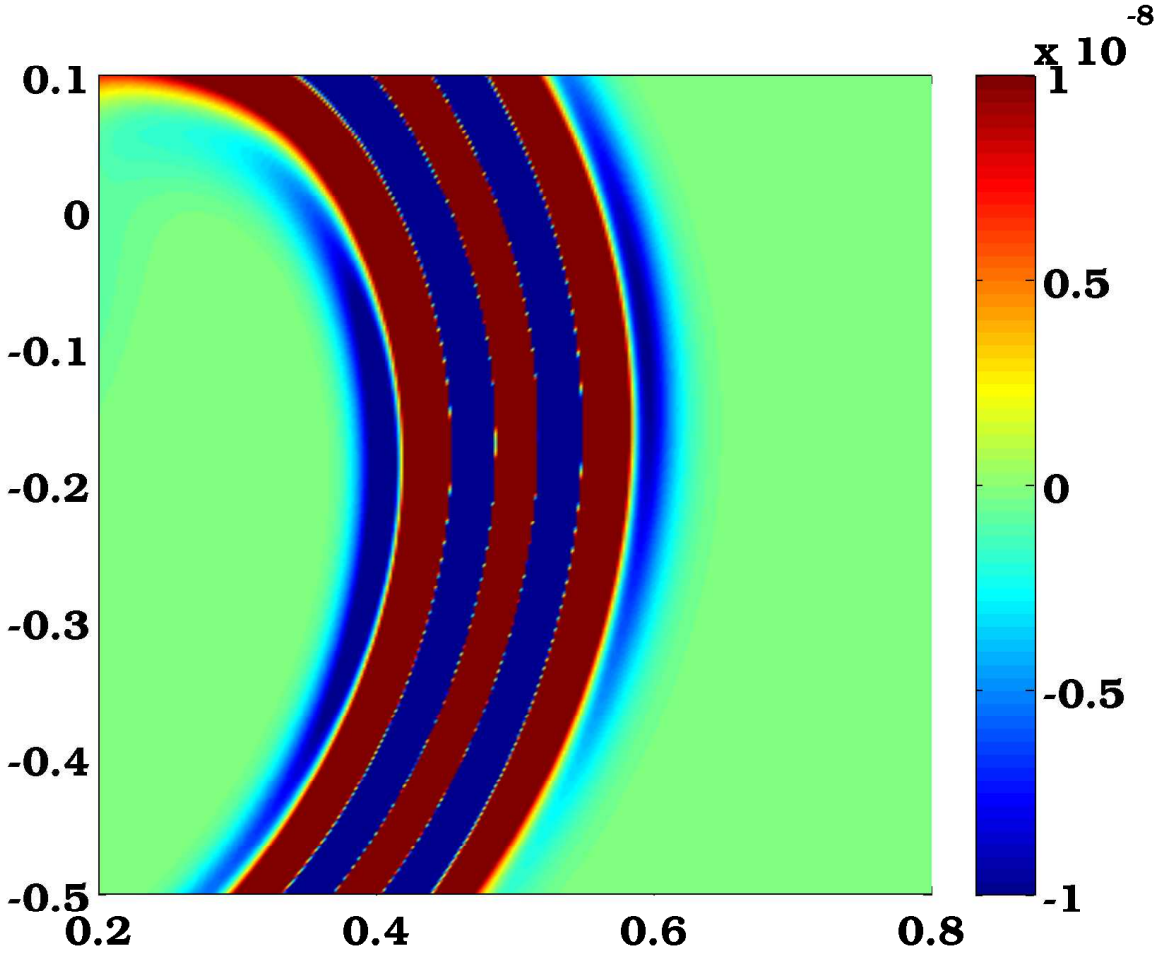


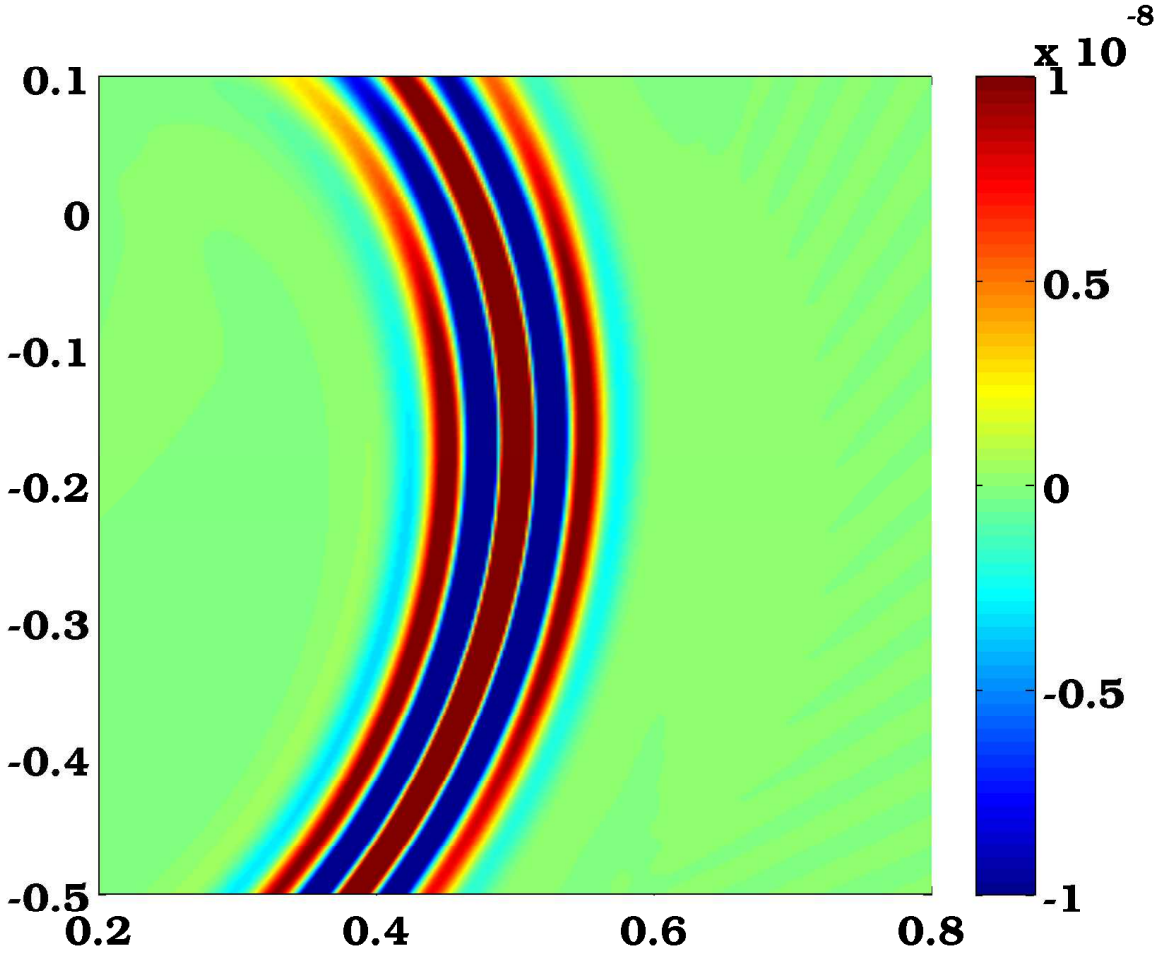
Verification of spatial errors

RK4 in time. Fourth order stencil.

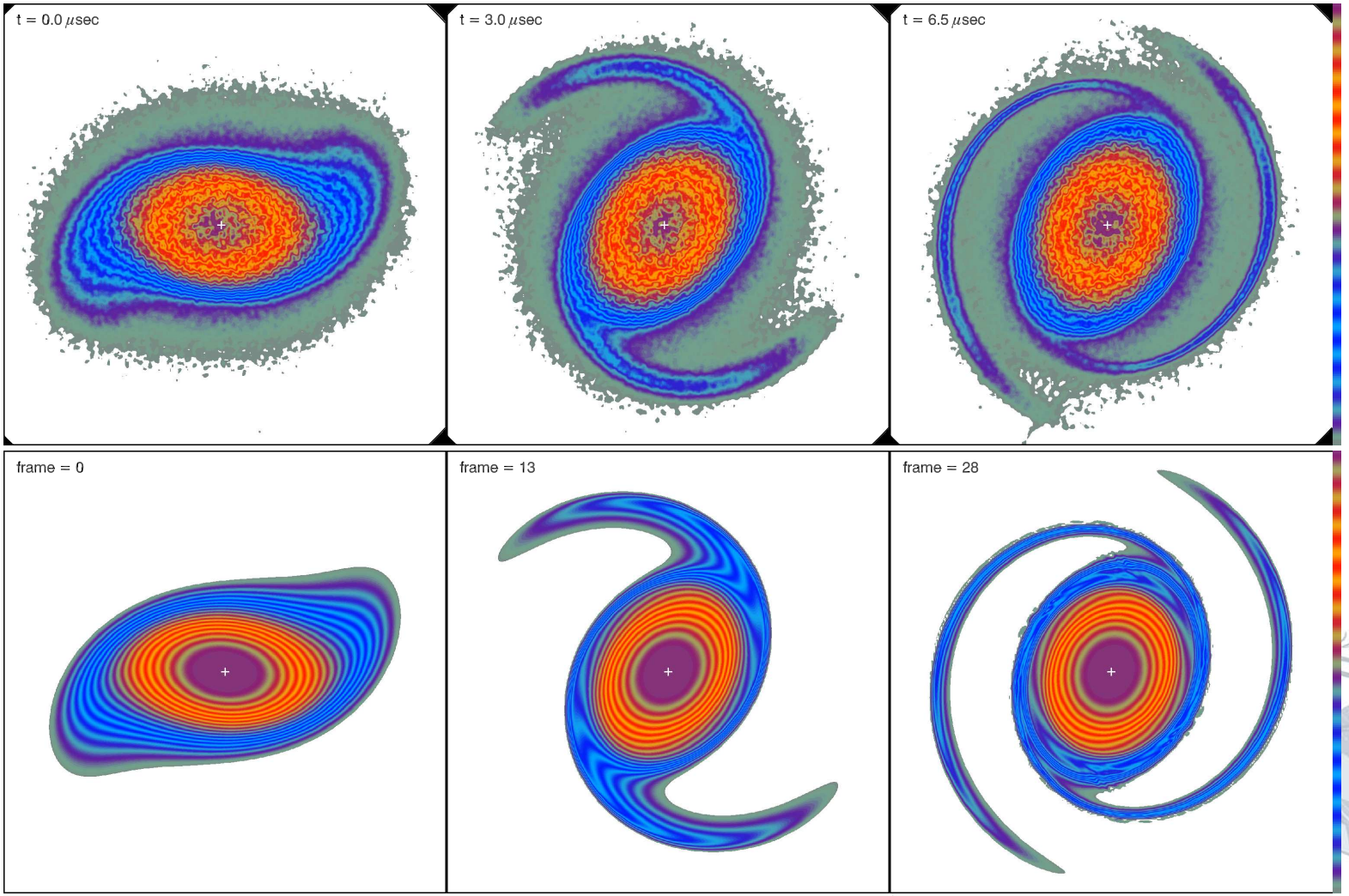


Verification of spatial errors

RK4 in time. Sixth order stencil.

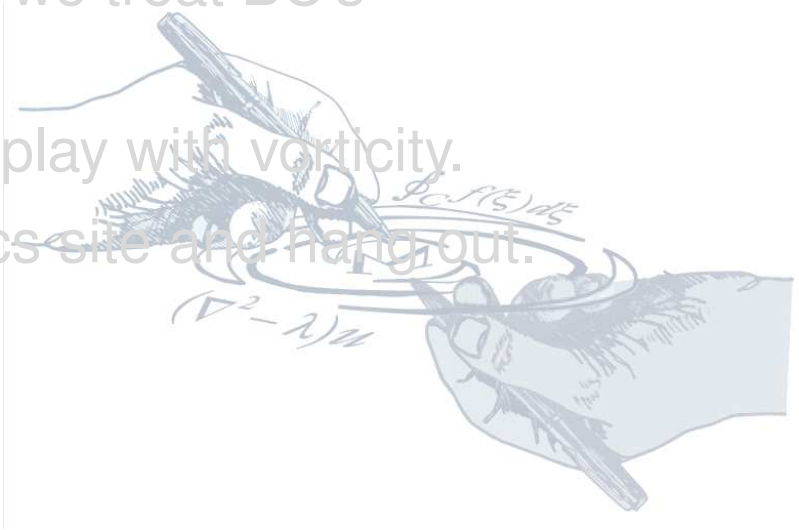


Physical experiments with electron vortices



Conclusions and future work

- Understanding convergence through the residual operator led to a new, better, general solver.
- The RHE technique works surprisingly well as a field interpolation method. Why? Would it make sense to refine RHE method in time? How should we treat BC's completely?
- Please visit the BlobFlow site and play with vorticity.
- Please visit the Meshfree Dynamics site and hang out.



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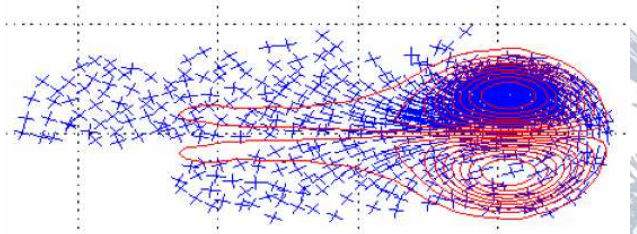

BlobFlow
Vorticity for all

www.math.udel.edu/∼rossi/BlobFlow

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Thank you!